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(54) **TRACKER MODULE AND
COMMUNICATION DEVICE**

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(57) **ABSTRACT**

An ET module is provided that includes a module laminate; first and second external output terminals disposed on the module laminate (90); and a voltage supply circuit disposed on the module laminate and configured to supply a power supply voltage V_{ETA} based on an envelope signal to a PA module via the first external output terminal, and to simultaneously supply a power supply voltage V_{ETB} based on an envelope signal to a PA module via the second external output terminal. The module laminate has first and second sides adjacent to each other, the first side and the second side forming at least a part of an outer shape of the module laminate in a plan view thereof. The first external output terminal is disposed adjacent to the first side, and the second external output terminal is disposed adjacent to the second side but not the first side.

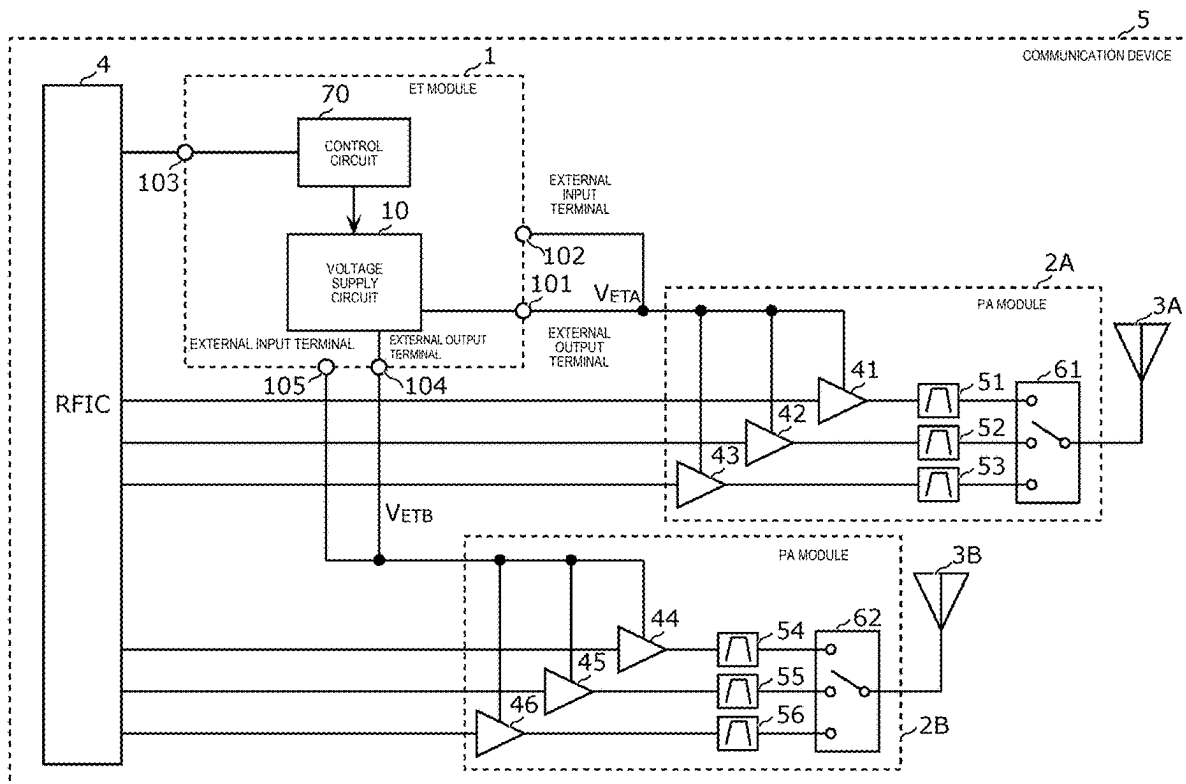


FIG. 1A

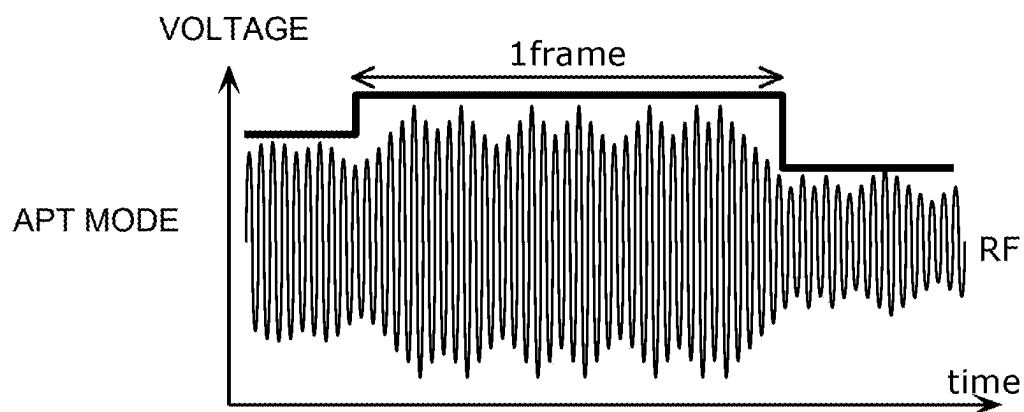


FIG. 1B

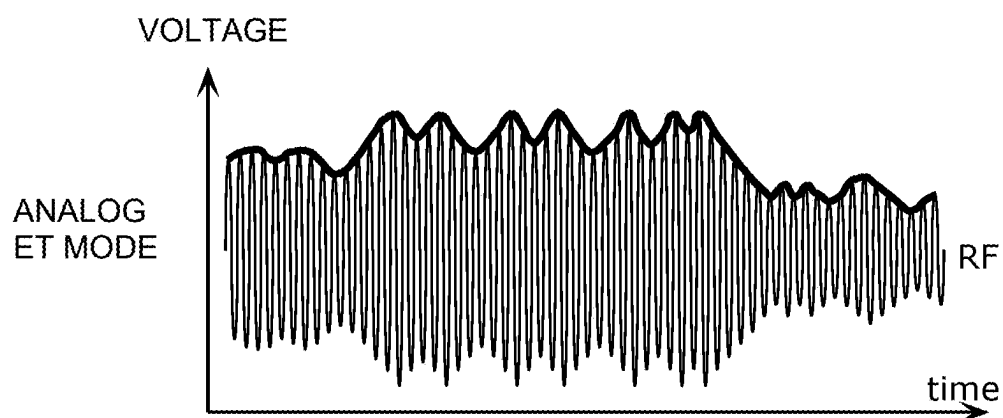


FIG. 1C

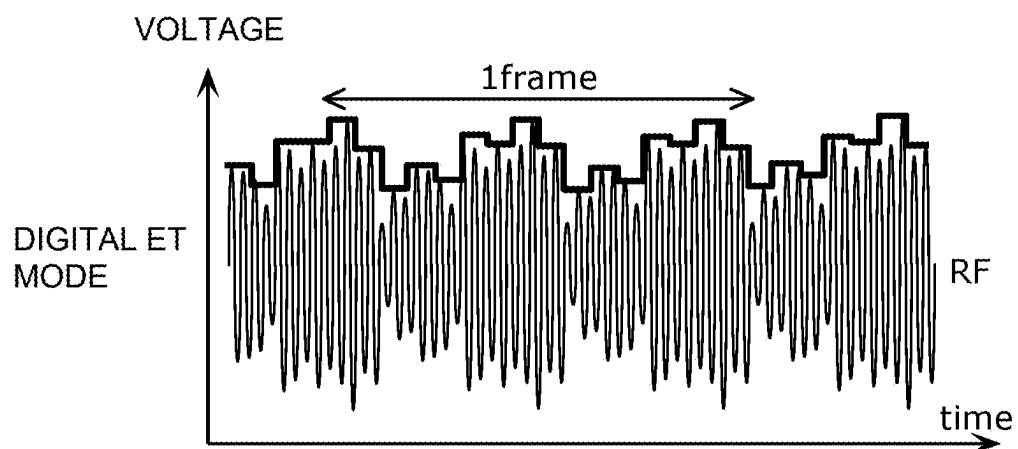


FIG. 2

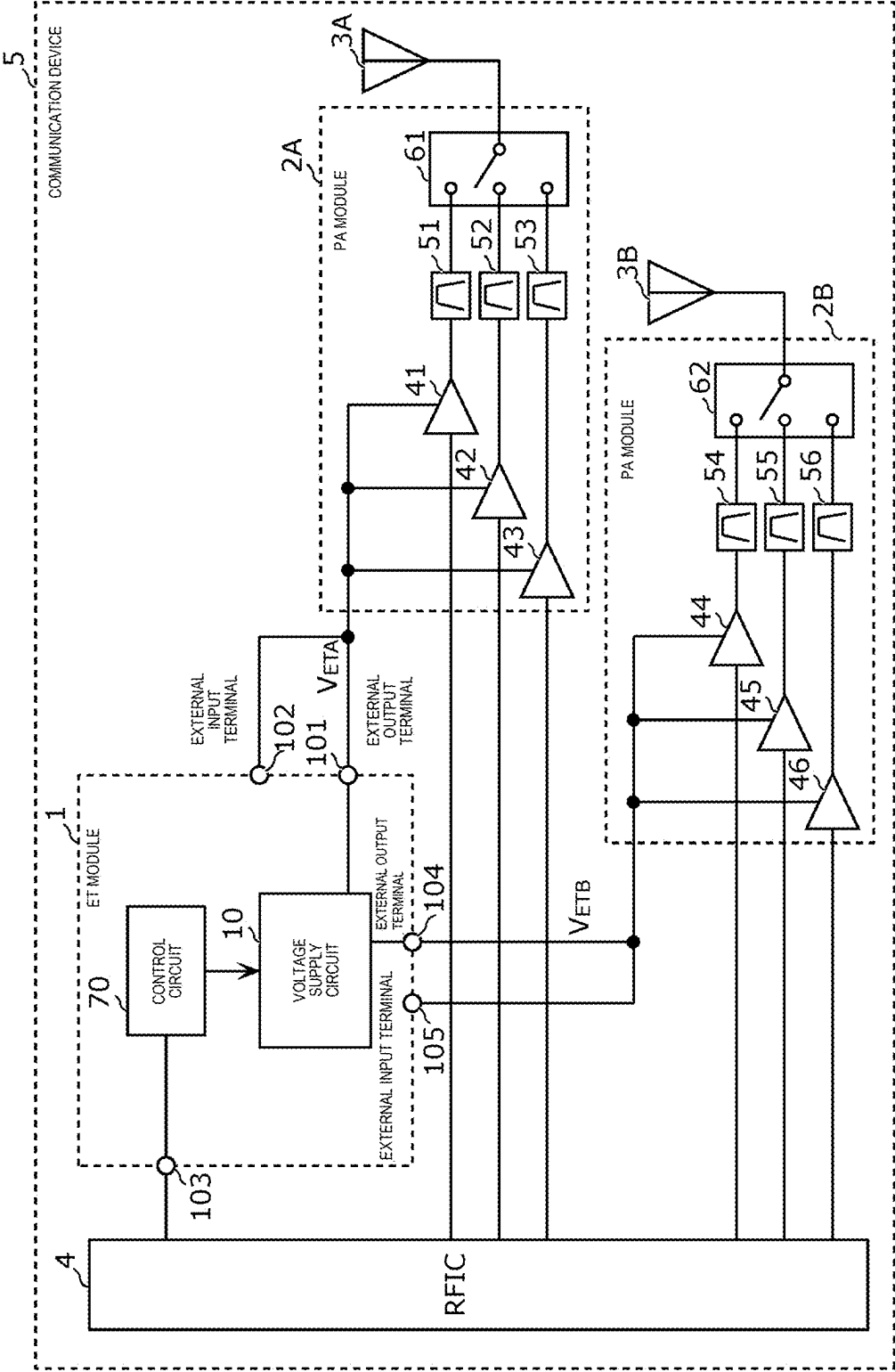


FIG. 3

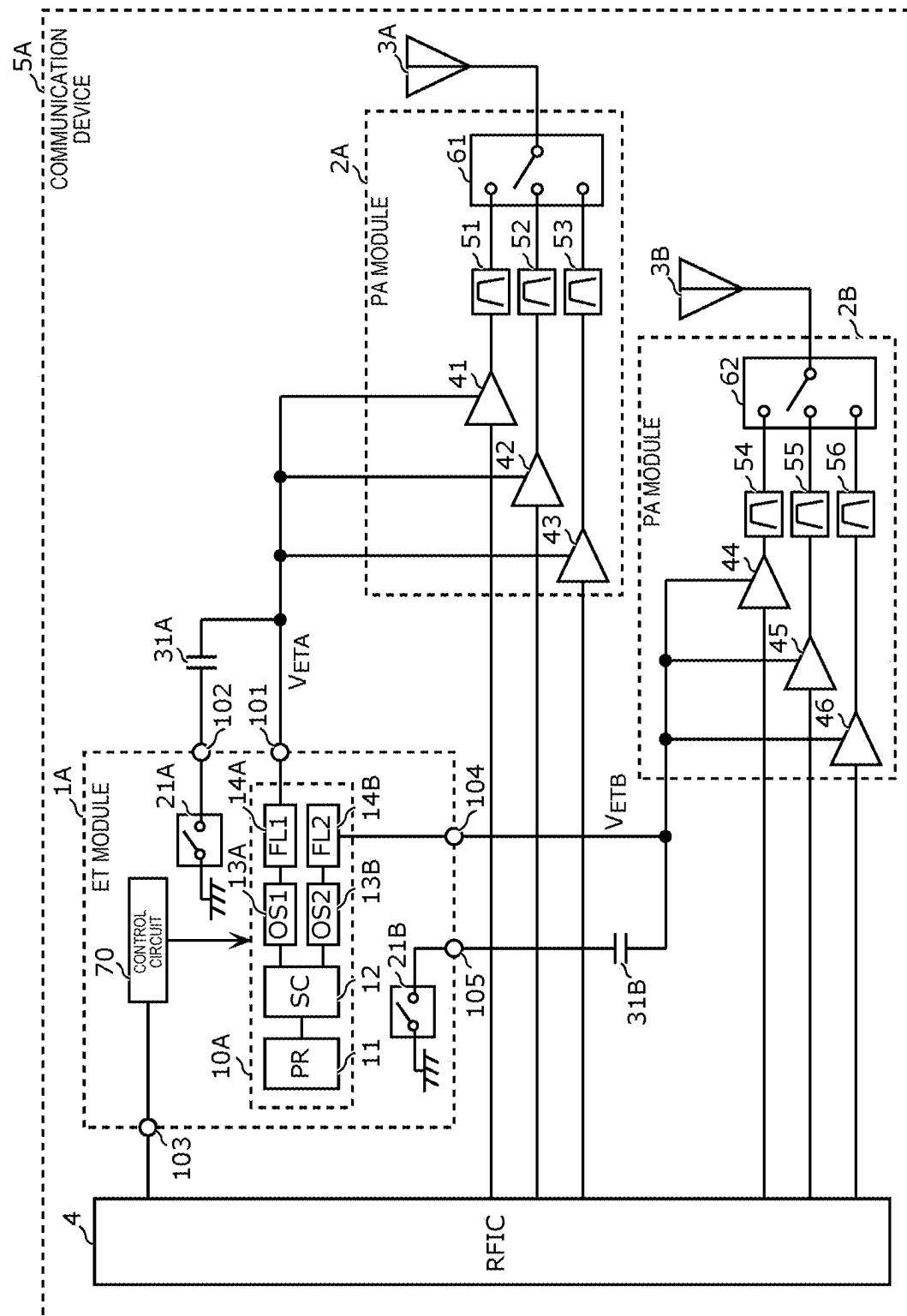


FIG. 4

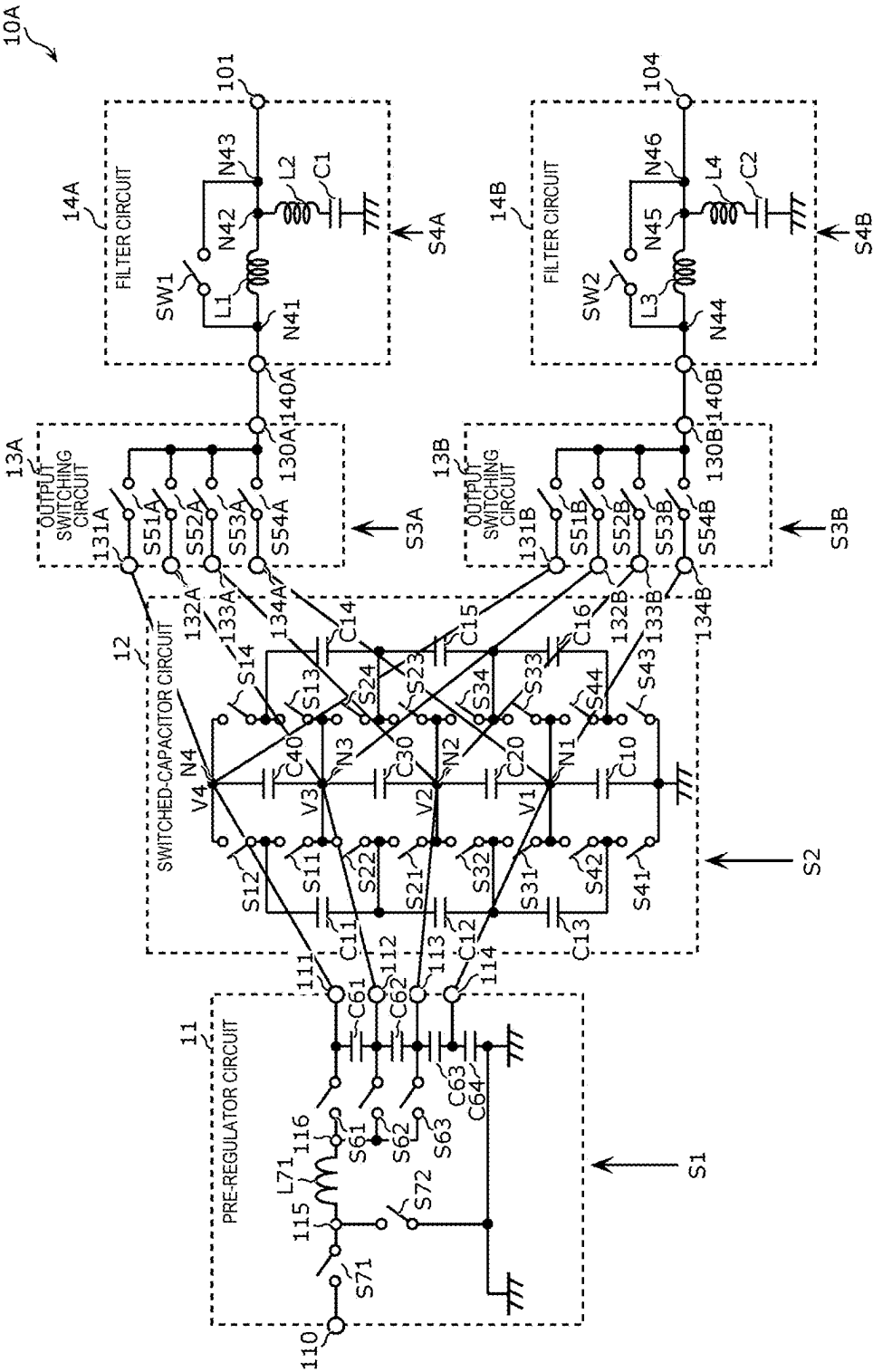


FIG. 5

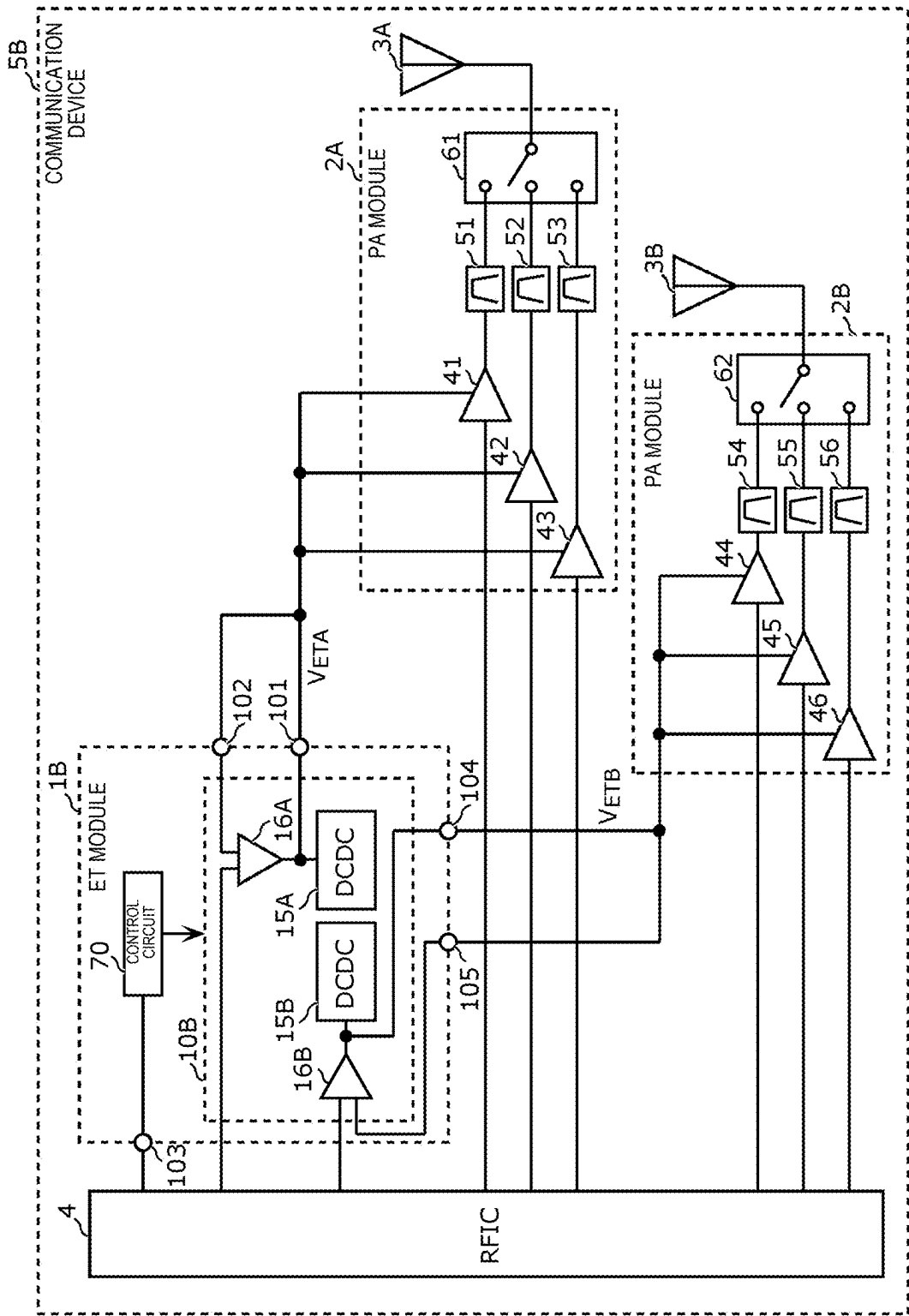


FIG. 6A

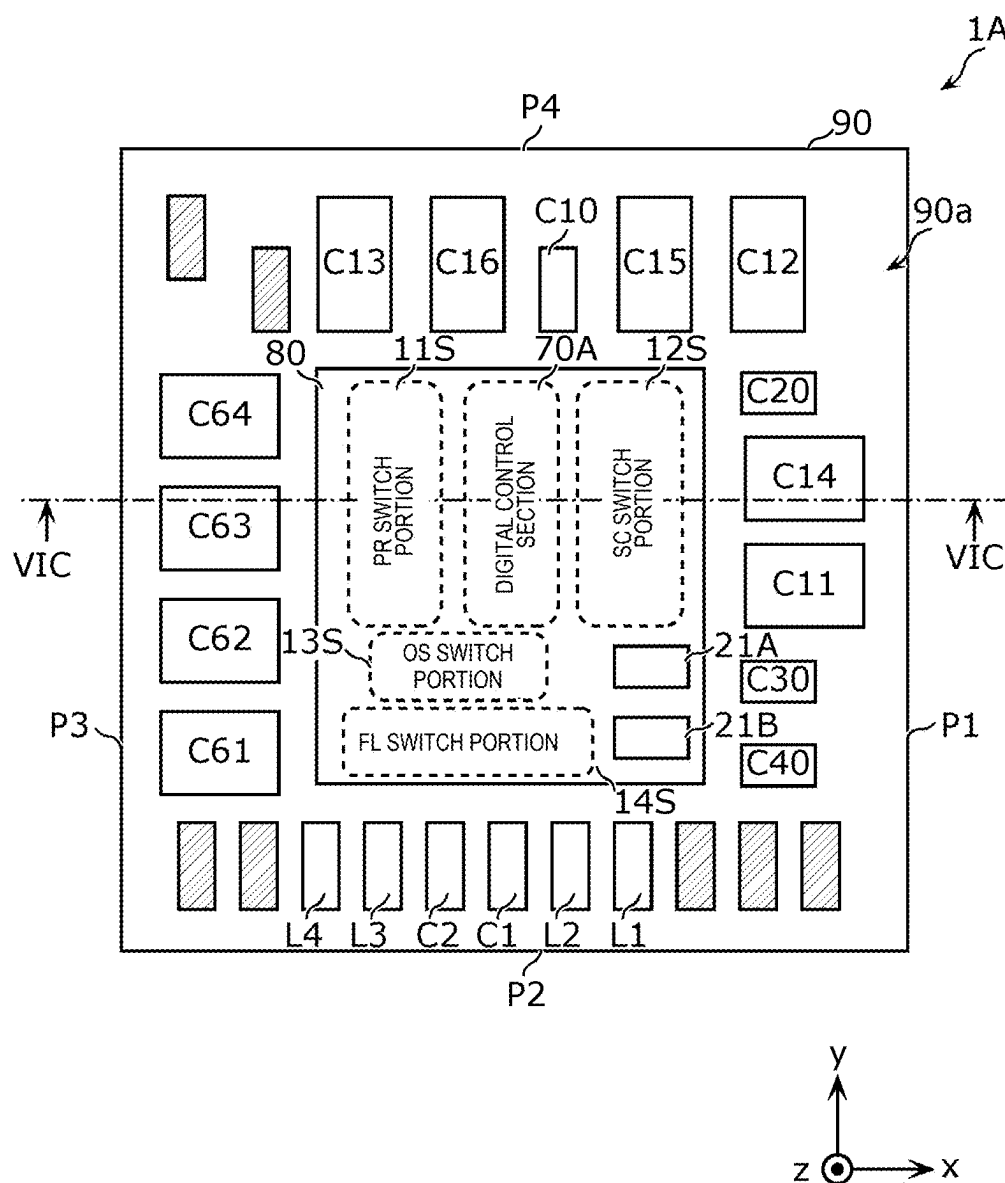


FIG. 6B

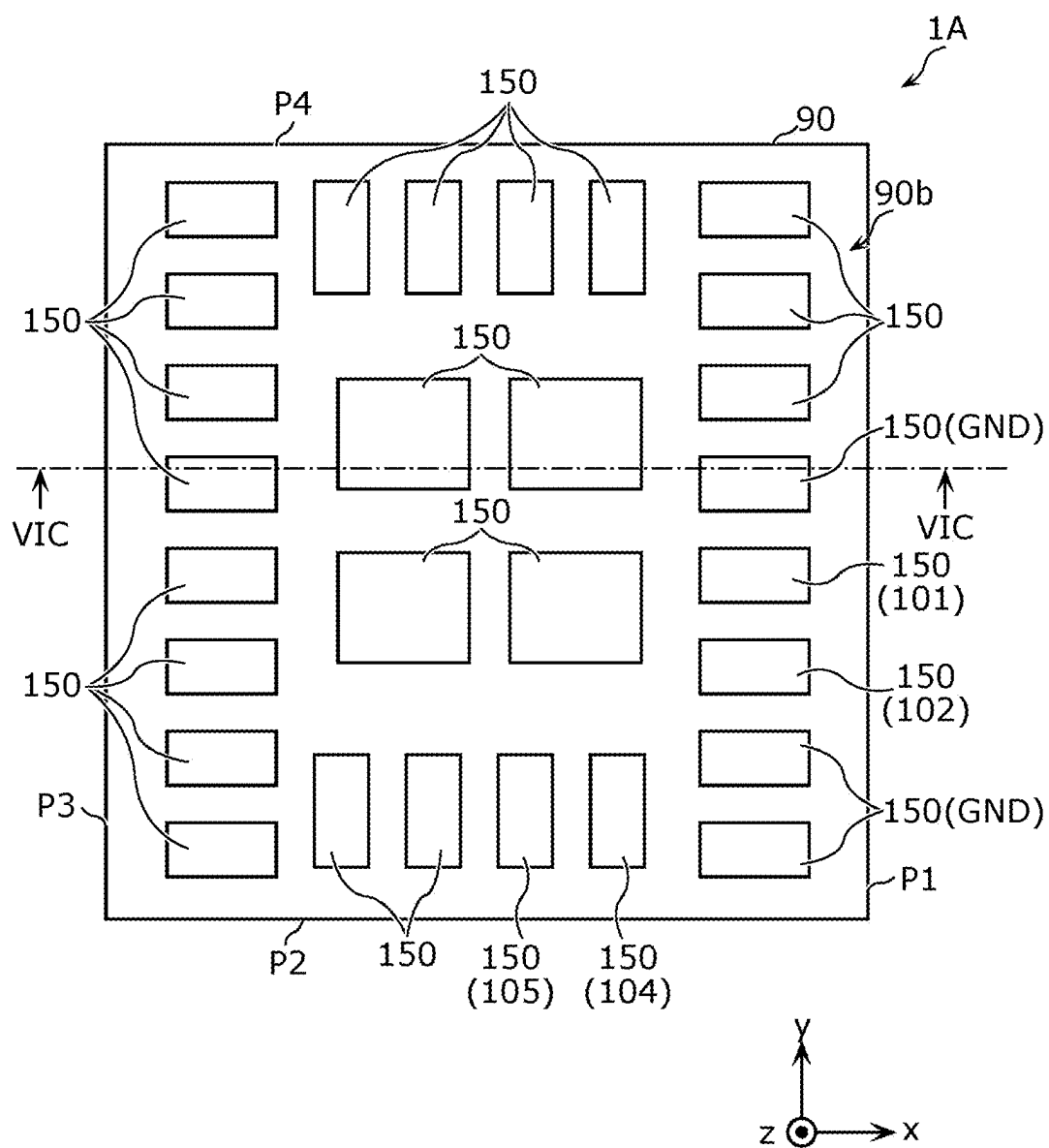


FIG. 6C

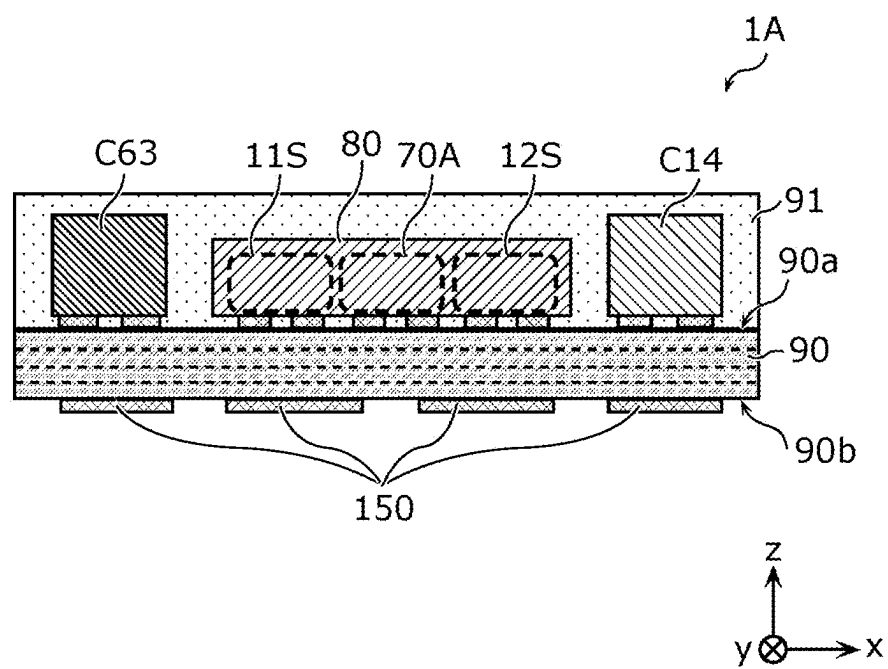
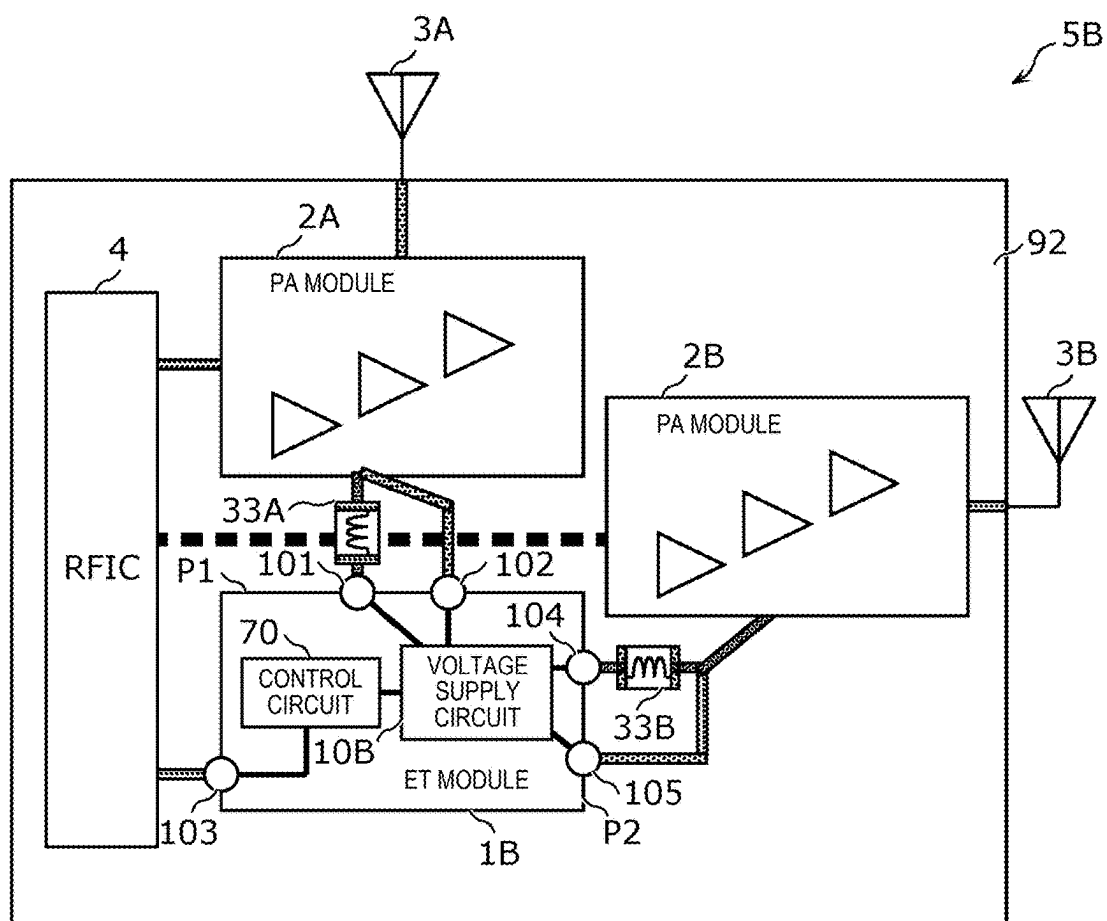


FIG. 8



TRACKER MODULE AND COMMUNICATION DEVICE

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of International Application No. PCT/JP2023/044470, filed Dec. 12, 2023, which claims priority to Japanese Patent Application No. 2022-206005, filed Dec. 22, 2022, the contents of each of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The exemplary aspects of the present disclosure relate to a tracker module and a communication device.

BACKGROUND

[0003] In recent years, a power added efficiency (hereinafter referred to as “efficiency”) has been improved by application of envelope tracking (ET: Envelope Tracking) to a power amplifier circuit. In ET systems, for example, a digital ET technology for supplying a plurality of discrete voltages has been disclosed.

[0004] For example, U.S. Pat. No. 9,755,672 discloses a supply modulator (e.g., a tracker circuit) that supplies a power supply voltage to a power amplifier circuit based on an envelope signal. The above-mentioned supply modulator includes a magnetic converter circuit (Magnetic Regulation Stage: Pre-regulator circuit) that converts a voltage; a switched-capacitor circuit (Switched-Capacitor Voltage Balance Stage) that generates, from the voltage, a plurality of voltages having different voltage levels; and an output switching circuit (Output Switching Stage) that selects and outputs at least one of the plurality of voltages. The magnetic converter circuit includes a switch and a power inductor, the switched-capacitor circuit includes a switch and a capacitor, and the output switching circuit includes a switch.

[0005] However, if a tracker circuit is modularized in an ET system, efficiency of the tracker circuit may deteriorate depending on an arrangement configuration of a voltage output path that outputs a voltage from the tracker circuit to a power amplifier circuit and a voltage input path that inputs the above voltage from the voltage output path to the tracker circuit.

SUMMARY OF THE INVENTION

[0006] In view of the foregoing, the exemplary aspects of the present disclosure provide a tracker module and a communication device that suppresses efficiency deterioration.

[0007] In an exemplary aspect, a tracker module is provided that includes a module laminate; a first external output terminal and a second external output terminal disposed on the module laminate; and a voltage supply circuit disposed on the module laminate and configured to supply a first voltage based on an envelope signal to a first power amplifier via the first external output terminal, and to simultaneously supply a second voltage based on an envelope signal to a second power amplifier via the second external output terminal. The module laminate has a first side and a second side adjacent to the first side. The first side and the second side form at least a part of an outer shape of the module laminate in a plan view of the module laminate. The first external output terminal is disposed adjacent to the first side,

and the second external output terminal is disposed adjacent to the second side but not the first side.

[0008] In another exemplary aspect, a tracker module is provided that includes a module laminate; a first external output terminal and a second external output terminal disposed on the module laminate; and a voltage supply circuit including at least one integrated circuit disposed on the module laminate. The at least one integrated circuit includes at least one switch included in a switched-capacitor circuit, at least one switch included in a first output switching circuit, and at least one switch included in a second output switching circuit. The switched-capacitor circuit is configured to generate a plurality of discrete voltages based on an input voltage and to output the generated plurality of discrete voltages to the first output switching circuit and the second output switching circuit. The first output switching circuit is configured to selectively output at least one discrete voltage of the generated plurality of discrete voltages to a first power amplifier via the first external output terminal, and the second output switching circuit is configured to selectively output at least one discrete voltage of the generated plurality of discrete voltages to a second power amplifier via the second external output terminal. The module laminate has a first side and a second side adjacent to the first side. The first side and the second side form at least a part of an outer shape of the module laminate in plan view of the module laminate. The first external output terminal is disposed adjacent to the first side, and the second external output terminal is disposed adjacent to the second side but not the first side.

[0009] According to the exemplary aspects of the present disclosure, a tracker module and a communication device are provided in which efficiency deterioration is suppressed.

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. 1A is a graph illustrating an example of a transition in a power supply voltage in an average power tracking (APT: Average Power Tracking) mode.

[0011] FIG. 1B is a graph illustrating an example of a transition in a power supply voltage in an analog ET mode.

[0012] FIG. 1C is a graph illustrating an example of a transition in a power supply voltage in a digital ET mode.

[0013] FIG. 2 is a circuit configuration diagram of a tracker module and a communication device according to an exemplary embodiment.

[0014] FIG. 3 is a circuit configuration diagram of a tracker module and a communication device according to Example 1.

[0015] FIG. 4 is a diagram of circuit configurations of a pre-regulator circuit, a switched-capacitor circuit, output switching circuits, and filter circuits according to Example 1.

[0016] FIG. 5 is a circuit configuration diagram of a tracker module and a communication device according to Example 2.

[0017] FIG. 6A is a plan view of the tracker module according to Example 1.

[0018] FIG. 6B is a plan view of the tracker module according to Example 1.

[0019] FIG. 6C is a cross-sectional view of the tracker module according to Example 1.

[0020] FIG. 7 is a plan view of the communication device according to Example 1.

[0021] FIG. 8 is a plan view of the communication device according to Example 2.

[0022] FIG. 9 is a plan view of a communication device according to a modification example.

DETAILED DESCRIPTION OF EMBODIMENTS

[0023] In the following, embodiments of the present disclosure are described in detail with reference to the drawings. All embodiments to be described below illustrate comprehensive and specific examples. Numeric values, shapes, materials, components, and an arrangement and a connection form of the components, which are described in the following embodiments, are merely examples and are not intended to limit the exemplary aspects of the present disclosure.

[0024] It is noted that each figure is a schematic diagram that is appropriately emphasized, omitted, or subjected to ratio adjustment in order to illustrate the exemplary aspects of the present disclosure. Each figure is not necessarily a strict illustration and may differ from an actual shape, a positional relationship, or a ratio. In each figure, an identical numeral is given to a substantially identical configuration, and an overlapping description may be omitted or simplified.

[0025] In each figure below, an x-axis and a y-axis are axes orthogonal to each other on a plane parallel to a main surface of a board. Specifically, if a board has a rectangular shape in plan view, the x-axis is parallel to a first side of the board, and the y-axis is parallel to a second side that is orthogonal to the first side of the board. In addition, a z-axis is an axis perpendicular to the main surface of the board, a positive direction of which indicates an upward direction and a negative direction of which indicates a downward direction.

[0026] In a circuit configuration of the present disclosure, the phrase “to be connected” includes a case where a connection terminal and/or a wiring conductor is directly connected, and a case in which a connection terminal and/or a wiring conductor is electrically connected via another circuit element. Moreover, the phrase “to be connected between A and B” can refer to a connection with both A and B between A and B.

[0027] In a component arrangement of the present disclosure, the phrase “a component disposed in or on a board” includes a component being disposed on a main surface of a board and a component being disposed in a board. Moreover, the phrase “a component being disposed on a main surface of a board” includes not only a component being disposed in contact with a main surface of a board, but also a component being disposed above the main surface without being in contact therewith (for example, a component being laminated on another component that is disposed in contact with a main surface). In addition, the phrase “a component being disposed on a main surface of a board” may include a component being disposed in a recessed portion formed on a main surface. The phrase “a component being disposed in a board” includes not only a component being encapsulated within a board but also an entire component being disposed between both main surfaces of a board, but a portion of the component not being covered with the board, and only a portion of the component being disposed in the board.

[0028] In the component arrangement of the present disclosure, the phrase “in plan view of a board” can mean viewing, from a z-axis positive side, an object or component orthographically projected on an xy-plane. The phrase “A overlapping with B in plan view” can mean at least a portion

of a region of A orthographically projected on the xy-plane overlaps with at least a portion of a region of B orthographically projected on the xy-plane. In addition, the phrase “A being disposed between B and C” can mean that at least one of a plurality of line segments connecting an arbitrary point in B and an arbitrary point in C passes through A.

[0029] In addition, in the component arrangement of the present disclosure, the phrase “A being disposed adjacent to B” represents that A and B are disposed in proximity, and specifically can mean that no other circuit component is present in a space in which A faces B. In other words, “A being disposed adjacent to B” can mean that none of a plurality of line segments reaching B, from an arbitrary point on a surface of A facing B, along a normal direction of the surface, passes through any circuit component other than A and B. Here, a circuit component can refer to a component including an active element and/or a passive element. More specifically, a circuit component includes an active component including a transistor or a diode, or the like, and a passive component including an inductor, a transformer, a capacitor, or a resistor, or the like, and does not include an electromechanical component including a terminal, a connector, or a wiring line, or the like.

[0030] In the exemplary aspects of the present disclosure, a “terminal” can refer to a point where a conductor within an element terminates. Moreover, it is noted that when impedance of a conductor between elements is sufficiently low, a terminal is construed not only as a single point but also as an arbitrary point on the conductor between the elements or the entire conductor.

[0031] In the component arrangement of the present disclosure, the phrase “component A being disposed in series on path B” can mean that both a signal input terminal and a signal output terminal of the component A are connected to a wiring line, an electrode, or a terminal that form the path B.

[0032] In addition and for purposes of this disclosure, terms that indicate a relationship between elements, such as “parallel” or “perpendicular”, and terms that indicate a shape of an element, such as “rectangular”, as well as a numerical range do not only indicate a strict meaning, but also may include a substantially equivalent range, for example, an error such as a few percent.

[0033] First, as a technology for amplifying radio frequency signals highly efficiently, a tracking mode is described that supplies a power amplifier with a power supply voltage to be dynamically adjusted over time based on the radio frequency signal. The tracking mode is a mode that dynamically adjusts a power supply voltage to be applied to the power amplifier. Although there are several types of tracking modes, here, an average power tracking (APT: Average Power Tracking) mode and an ET (ET: Envelope Tracking mode (including an analog ET mode and a digital ET mode) are described with reference to FIGS. 1A to 1C. In FIGS. 1A to 1C, a horizontal axis represents time and a vertical axis represents a voltage. In addition, a thick solid line represents a power supply voltage and a thin solid line (waveform) represents a modulated signal.

[0034] FIG. 1A is a graph illustrating an example of a transition in a power supply voltage in the APT mode. In the APT mode, a power supply voltage is fluctuated to a plurality of discrete voltage levels in units of one frame based on average power. As a result, a power supply voltage signal forms a rectangular wave.

[0035] According to an exemplary aspect, a frame can refer to a unit that forms a radio frequency signal (e.g., a modulated signal). For example, in 5G NR (5th Generation New Radio) and LTE (Long Term Evolution), a frame includes 10 subframes, each subframe including a plurality of slots, each slot being composed of a plurality of symbols. A subframe has a length of 1 ms, and a frame has a length of 10 ms.

[0036] It is noted that a mode of fluctuating a voltage level in units of one frame or larger based on the average power is referred to as the APT mode, and is distinguished from a mode that fluctuates the voltage level in units smaller than one frame (for example, a subframe, a slot, or a symbol). For example, a mode that fluctuates the voltage level in symbol units is referred to as a symbol power tracking (SPT: Symbol Power Tracking) mode and is distinguished from the APT mode.

[0037] FIG. 1B is a graph illustrating an example of a transition in a power supply voltage in an analog ET mode. In the analog ET mode, an envelope of the modulated signal is tracked by continuously fluctuating the power supply voltage based on an envelope signal.

[0038] An envelope signal is a signal representing an envelope of a modulated signal. An envelope value is expressed by, for example, a root square of ($I^2 + Q^2$). Here, (I, Q) represents a constellation point. A constellation point refers to a point that represents a signal modulated by digital modulation on a constellation diagram. (I, Q) is determined by, for example, BBIC (Baseband Integrated Circuit) based on, for example, transmission information.

[0039] FIG. 1C is a graph illustrating an example of a transition in a power supply voltage in a digital ET mode. In the digital ET mode, an envelope of a modulated signal is tracked by fluctuating the power supply voltage to a plurality of discrete voltage levels in one frame based on an envelope signal. As a result, the power supply voltage signal forms a rectangular wave.

Exemplary Embodiments

[1. Circuit Configuration of ET Module 1 and Communication Device 5]

[0040] An ET module 1 and a communication device 5 according to the present embodiment are described with reference to FIG. 2.

[0041] FIG. 2 is a circuit configuration diagram of the ET module 1 and the communication device 5 according to the present embodiment. The communication device 5 according to the present embodiment corresponds to a user terminal (UE User Equipment) in a cellular network, and is typically a mobile phone, a smart phone, a tablet computer, a wearable device, or the like. It is noted that the communication device 5 may be an IoT (Internet of Things) sensor device, a medical/healthcare device, an automobile, an unmanned aerial vehicle (UAV: Unmanned Aerial Vehicle) (a so-called drone), and an automated guided vehicle (AGV: Automated Guided Vehicle).

[0042] First, a circuit configuration of the communication device 5 is described. As illustrated in FIG. 2, the communication device 5 according to the present embodiment includes the ET module 1, PA modules 2A and 2B, antennae 3A and 3B, and an RFIC (Radio Frequency Integrated Circuit) 4.

[0043] The antenna 3A is connected to the PA module 2A and transmits a radio frequency signal output from the PA module 2A. The antenna 3B is connected to the PA module 2B and transmits a radio frequency signal output from the PA module 2B.

[0044] The RFIC 4 is an example of a signal processing circuit that processes radio frequency signals. The RFIC 4 has a control section that controls the PA modules 2A and 2B. Specifically, the RFIC 4 performs signal processing on a transmission signal input from the BBIC (not illustrated) by up-conversion, or the like, and outputs a radio frequency transmission signal generated by the signal processing to the PA modules 2A and 2B. In addition, the RFIC 4 outputs a digital control signal that controls the ET module 1 to the ET module 1. It is noted that some or all of capabilities of the RFIC 4 as the control section may be implemented externally to the RFIC 4, and may be implemented, for example, in the BBIC, the PA modules 2A and 2B, and the ET module 1.

[0045] The ET module 1 supplies a power supply voltage V_{ETA} to the PA module 2A and supplies a power supply voltage V_{ETB} to the PA module 2B. The power supply voltage V_{ETA} and the power supply voltage V_{ETB} are examples of voltages based on envelope signals. The ET module 1 can supply the power supply voltage V_{ETA} to the PA module 2A and supply the power supply voltage V_{ETB} to the PA module 2B in at least any of the APT mode, the analog ET mode, and the digital ET mode illustrated in FIGS. 1A to 1C.

[0046] As illustrated in FIG. 2, the ET module 1 includes a voltage supply circuit 10, a control circuit 70, external output terminals 101 and 104, external input terminals 102 and 105, and control signal terminal 103.

[0047] The voltage supply circuit 10 is configured to supply the power supply voltage V_{ETA} based on the envelope signal to the PA module 2A by way of the external output terminal 101, and to supply the power supply voltage V_{ETB} based on the envelope signal to the PA module 2B by way of an external output terminal 104. Specifically, the voltage supply circuit 10 receives a control signal output from the control circuit 70 to generate the power supply voltage V_{ETA} , and outputs the power supply voltage V_{ETA} to the external output terminal 101. In addition, the voltage supply circuit 10 receives a control signal output from the control circuit 70 to generate the power supply voltage V_{ETB} , and outputs the power supply voltage V_{ETB} to the external output terminal 104. The voltage supply circuit 10 is, for example, a DC-DC converter that steps up or down a pressure of magnetic energy generated by an inductor of the DC-DC converter and outputs the magnetic energy, as the power supply voltage V_{ETA} and the power supply voltage V_{ETB} . It is noted that the inductor that generates the magnetic energy may be located externally to the ET module 1 in an exemplary aspect.

[0048] The external output terminal 101 is an example of a first external output terminal that can be connected externally and is connected to the voltage supply circuit 10 and the PA module 2A. The power supply voltage V_{ETA} , which is generated by the voltage supply circuit 10, is applied to the external output terminal 101.

[0049] The external output terminal 104 is an example of a second external output terminal that can be connected externally and is connected to the voltage supply circuit 10 and the PA module 2B. The power supply voltage V_{ETB} ,

which is generated by the voltage supply circuit 10, is applied to the external output terminal 104.

[0050] The external input terminal 102 is an example of a first external input terminal that can be connected externally and is connected to a first path connecting the external output terminal 101 and the PA module 2A. In operation, when the analog ET mode is applied to the ET module 1, the external input terminal 102 is configured to function as a terminal for feeding back the power supply voltage V_{ETA} on the first path to the ET module 1, and is connected to the voltage supply circuit 10 or the control circuit 70 within the ET module 1. In addition, when the APT mode is applied to the ET module 1, the external input terminal 102 is configured to function as a terminal for stabilizing the power supply voltage V_{ETA} on the first path, and is connected to a capacitor, for example.

[0051] The external input terminal 105 is an example of the second external input terminal that can be connected externally and is connected to a second path connecting the external output terminal 104 and the PA module 2B. In operation, when the analog ET mode is applied to the ET module 1, the external input terminal 105 is configured to function as a terminal for feeding back the power supply voltage V_{ETB} on the second path to the ET module 1, and is connected to the voltage supply circuit 10 or the control circuit 70 within the ET module 1. In addition, when the APT mode is applied to the ET module 1, the external input terminal 105 is configured to function as a terminal for stabilizing the power supply voltage V_{ETB} on the second path, and is connected to a capacitor, for example.

[0052] The control circuit 70 is an example of a control section of the ET module 1 and is connected to the control signal terminal 103 and the voltage supply circuit 10. The control circuit 70 controls the voltage supply circuit 10 to output the power supply voltage V_{ETA} and the power supply voltage V_{ETB} from the voltage supply circuit 10, based on an envelope signal input from the RFIC 4 via the control signal terminal 103, target power supply voltage information of the PA modules 2A and 2B, and a feedback power supply voltage input via the external input terminals 102 and 105.

[0053] The PA module 2A receives supply of the power supply voltage V_{ETA} from the ET module 1, amplifies a radio frequency transmission signal input from the RFIC 4, and outputs the amplified radio frequency transmission signal to the antenna 3A. The PA module 2A includes power amplifiers 41, 42, and 43, filters 51, 52, and 53, and a switch 61.

[0054] Each of the power amplifiers 41 to 43 is an example of a first power amplifier and is connected between the ET module 1 and the antenna 3A. Specifically, a supply terminal of the power amplifier 41 is connected to the ET module 1, a radio frequency input terminal is connected to the RFIC 4, and a radio frequency output terminal is connected to the antenna 3A via the filter 51 and the switch 61. A supply terminal of the power amplifier 42 is connected to the ET module 1, a radio frequency input terminal is connected to the RFIC 4, and a radio frequency output terminal is connected to the antenna 3A via the filter 52 and the switch 61. A supply terminal of the power amplifier 43 is connected to the ET module 1, a radio frequency input terminal is connected to the RFIC 4, and a radio frequency output terminal is connected to the antenna 3A via the filter 53 and the switch 61. The power amplifier 41 can amplify a radio frequency signal of band A, the power amplifier 42 can

amplify a radio frequency signal of band B, and the power amplifier 43 can amplify a radio frequency signal of band C.

[0055] The filter 51 includes the band A as a pass band, for example. The filter 52 includes the band B as a pass band, for example. The filter 53 includes the band C as a pass band, for example.

[0056] The switch 61 has a common terminal and three selection terminals, and switches connection between the common terminal and any one of the three selection terminals. The common terminal of the switch 61 is connected to the antenna 3A, a first selection terminal is connected to the filter 51, a second selection terminal is connected to the filter 52, and a third selection terminal is connected to the filter 53.

[0057] According to a configuration of the PA module 2A described above, the PA module 2A can amplify a radio frequency signal of any one of the band A, the band B, and the band C, and output the amplified radio frequency signal to the antenna 3A.

[0058] A node on the first path connecting each of the supply terminals of the power amplifiers 41 to 43 and the external output terminal 101 is connected to the external input terminal 102.

[0059] The PA module 2B receives supply of the power supply voltage V_{ETB} from the ET module 1, amplifies a radio frequency transmission signal input from the RFIC 4, and outputs the amplified radio frequency transmission signal to the antenna 3B. The PA module 2B includes power amplifiers 44, 45 and 46, filters 54, 55, and 56, and a switch 62.

[0060] Each of the power amplifiers 44 to 46 is an example of a second power amplifier and is connected between the ET module 1 and the antenna 3B. Specifically, a supply terminal of the power amplifier 44 is connected to the ET module 1, a radio frequency input terminal is connected to the RFIC 4, and a radio frequency output terminal is connected to the antenna 3B via the filter 54 and the switch 62. A supply terminal of the power amplifier 45 is connected to the ET module 1, a radio frequency input terminal is connected to the RFIC 4, and a radio frequency output terminal is connected to the antenna 3B via the filter 55 and the switch 62. A supply terminal of the power amplifier 46 is connected to the ET module 1, a radio frequency input terminal is connected to the RFIC 4, and a radio frequency output terminal is connected to the antenna 3B via the filter 56 and the switch 62. The power amplifier 44 can amplify a radio frequency signal of band D, the power amplifier 45 can amplify a radio frequency signal of band E, and the power amplifier 46 can amplify a radio frequency signal of band F.

[0061] The filter 54 includes the band D as a pass band, for example. The filter 55 includes the band E as a pass band, for example. The filter 56 includes the band F as a pass band, for example.

[0062] The switch 62 has a common terminal and three selection terminals, and switches connection between the common terminal and any one of the three selection terminals. The common terminal of the switch 62 is connected to the antenna 3B, a first selection terminal is connected to the filter 54, a second selection terminal is connected to the filter 55, and a third selection terminal is connected to the filter 56.

[0063] According to a configuration of the PA module 2B described above, the PA module 2B can amplify a radio frequency signal of any one of the band D, the band E, and the band F, and output the amplified radio frequency signal to the antenna 3B.

[0064] A node on the second path connecting each of the supply terminals of the power amplifiers 44 to 46 and the external output terminal 104 is connected to the external input terminal 105.

[0065] It is noted that each of the band A to the band F is a frequency band for a communication system that is built using a radio access technology (RAT: Radio Access Technology), and is predefined by a standardizing body (for example, 3GPP® (3rd Generation Partnership Project) and IEEE (Institute of Electrical and Electronics Engineers), or the like). Examples of communication systems include a 5G NR (5th Generation New Radio) system, an LTE (Long Term Evolution) system, and a WLAN (Wireless Local Area Network) system, or the like.

[0066] According to the above-described configuration, the communication device 5 can simultaneously transmit a radio frequency signal of any of the band A to the band C and a radio frequency signal of any of the band D to the band F. In order to cope with this, the ET module 1 is configured to supply the power supply voltage V_{ETA} (e.g., a first voltage) based on the envelope signal to the PA module 2A via the external output terminal 101, and simultaneously supply the power supply voltage V_{ETB} (e.g., a second voltage) based on the envelope signal to the PA module 2B via the external output terminal 104.

[2. Circuit Configuration of ET Module 1A and Communication Device 5A According to Example 1]

[0067] FIG. 3 is a circuit configuration diagram of an ET module 1A and the communication device 5A according to Example 1. The ET module 1A and the communication device 5A according to this Example are examples of the ET module 1 and the communication device 5 according to the present embodiment and illustrate a specific circuit configuration of the ET module 1.

[0068] The communication device 5A according to this Example includes the ET module 1A, the PA modules 2A and 2B, capacitors 31A and 31B, antennae 3A and 3B, and the RFIC 4. As compared to the communication device 5 according to the embodiment, the communication device 5A according to this Example differs in that a configuration of the ET module 1A as well as the capacitors 31A and 31B are added. Therefore, in the following, the communication device 5A according to this Example is described with a focus on configurations of the ET module 1A and the capacitors 31A and 31B.

[0069] The ET module 1A is an example of a tracker module, and can supply the power supply voltage V_{ETA} , which is one of a plurality of discrete voltages, to the PA module 2A, and supply the power supply voltage V_{ETB} , which is one of the plurality of discrete voltages, to the PA module 2B. The digital ET mode and the APT mode can be used as a tracking mode, but the tracking mode is not limited thereto. As illustrated in FIG. 3, the ET module 1A includes a voltage supply circuit 10A, the control circuit 70, switches 21A and 21B, the external output terminals 101 and 104, the external input terminals 102 and 105, and the control signal terminal 103. As compared to the ET module 1 according to the embodiment, the ET module 1A according to this Example differs in that a configuration of the voltage supply circuit 10A and the switches 21A and 21B are added. In the following, the ET module 1A according to this Example is

described, omitting the same configurations as those of the ET module 1 according to the embodiment and focusing on different configurations.

[0070] The voltage supply circuit 10A includes a pre-regulator circuit 11, a switched-capacitor circuit 12, output switching circuits 13A and 13B, and filter circuits 14A and 14B.

[0071] The pre-regulator circuit 11 is an example of a converter circuit and includes a power inductor and a switch. A power inductor is an inductor used to step up and/or down a direct current (DC: Direct Current) voltage. The power inductor is connected in series to a direct current path. It is noted that the power inductor may be connected (e.g., disposed in parallel) between the direct current path and a ground according to an exemplary aspect. The pre-regulator circuit 11 can use the power inductor to convert an input voltage into a third voltage. In some cases, such a pre-regulator circuit 11 may be referred to as a magnetic regulator or a DC-DC converter.

[0072] The switched-capacitor circuit 12 includes a plurality of capacitors and a plurality of switches, and can be configured to generate a plurality of discrete voltages from input voltages from the pre-regulator circuit 11. The switched-capacitor circuit 12 may be referred to as a switched-capacitor voltage balancer (Switched-Capacitor Voltage Balancer).

[0073] The output switching circuit 13A is an example of a first output switching circuit and is configured to selectively output at least one of the plurality of discrete voltages generated by the switched-capacitor circuit 12 to the PA module 2A via the external output terminal 101. The output switching circuit 13A is controlled based on a digital control signal.

[0074] The output switching circuit 13B is an example of a second output switching circuit and is configured to selectively output at least one of the plurality of discrete voltages generated by the switched-capacitor circuit 12 to the PA module 2B via the external output terminal 104. The output switching circuit 13B is controlled based on a digital control signal.

[0075] The filter circuit 14A can attenuate noise from the plurality of discrete voltages supplied to the PA module 2A. The filter circuit 14B can attenuate noise from the plurality of discrete voltages supplied to the PA module 2B. The filter circuits 14A and 14B may be referred to as pulse shaping filters or transition shaping filters.

[0076] It is noted that the voltage supply circuit 10A may omit the filter circuits 14A and 14B in an exemplary aspect. In addition, any combination of the pre-regulator circuit 11, the switched-capacitor circuit 12, the output switching circuits 13A and 13B, and the filter circuits 14A and 14B may be integrated into a single circuit. Alternatively, the voltage supply circuit 10A may include a plurality of voltage supply circuits, instead of the pre-regulator circuit 11 and the switched-capacitor circuit 12. In this case, the output switching circuits 13A and 13B may be each configured to select at least one of the plurality of voltage supply circuits.

[0077] The switch 21A is an example of a first switch circuit and is connected between the external input terminal 102 and the ground to switch between connection and disconnection between the external input terminal 102 and the ground. The switch 21B is connected between the external input terminal 105 and the ground, and switches

between connection and disconnection between the external input terminal 105 and the ground.

[0078] The capacitor 31A is disposed in series between the external input terminal 102 and a node on the first path. The capacitor 31B is disposed in series between the external input terminal 105 and a node on the second path.

[0079] In operation, when the digital ET mode is applied to the ET module 1A, for example, the switches 21A and 21B are in a non-conductive state, so that a radio frequency signal of a wide channel band width (wide modulation band width) can be amplified highly efficiently in the PA modules 2A and 2B. In addition, when the APT mode is applied to the ET module 1A, for example, the switches 21A and 21B are in a conductive state, so that the capacitors 31A and 31B can be configured to function as bypass condensers, and voltage levels of the power supply voltage V_{ETA} and the power supply voltage V_{ETB} can be stabilized.

[0080] Next, circuit configuration of the pre-regulator circuit 11, the switched-capacitor circuit 12, the output switching circuits 13A and 13B, and the filter circuits 14A and 14B, which are included in the voltage supply circuit 10A, are described with reference to FIG. 4.

[0081] FIG. 4 is a diagram of circuit configurations of the pre-regulator circuit 11, the switched-capacitor circuit 12, the output switching circuits 13A and 13B, and filter circuits 14A and 14B according to Example 1.

[0082] It is noted that FIG. 4 is an exemplary circuit configuration, and that the pre-regulator circuit 11, the switched-capacitor circuit 12, the output switching circuits 13A and 13B, and the filter circuits 14A and 14B may be implemented using any of a wide variety of circuit implementations or circuit technologies. Therefore, a description of each of the circuits to be provided below should not be interpreted in a limited manner.

[2.1 Circuit Configuration of Switched-Capacitor Circuit 12]

[0083] As illustrated in FIG. 4, the switched-capacitor circuit 12 includes capacitors C11 to C16, capacitors C10, C20, C30, and C40, and switches S11 to S14, S21 to S24, S31 to S34, and S41 to S44. Energy and electric charges are input from the pre-regulator circuit 11 to the switched-capacitor circuit 12 at nodes N1 to N4 and drawn from the switched-capacitor circuit 12 to the output switching circuits 13A and 13B at the nodes N1 to N4.

[0084] Each of the capacitors C11 to C16 is configured to function as a flying capacitor (which may be referred to as a transfer capacitor) in an exemplary aspect. That is, each of the capacitors C11 to C16 are used to step up or down input voltages supplied from the pre-regulator circuit 11. More specifically, the capacitors C11 to C16 transfer electric charges between the capacitors C11 to C16 and the nodes N1 to N4 so that voltages V1 to V4 (voltages with respect to a ground potential), which satisfy $V1:V2:V3:V4=1:2:3:4$ at the four nodes N1 to N4, are maintained. These voltages V1 to V4 correspond to a plurality of discrete voltages.

[0085] The capacitor C11 has two electrodes. One of the two electrodes of the capacitor C11 is connected to one end of the switch S11 and one end of the switch S12. The other of the two electrodes of the capacitor C11 is connected to one end of the switch S21 and one end of the switch S22.

[0086] The capacitor C12 has two electrodes. One of the two electrodes of the capacitor C12 is connected to the one end of the switch S21 and the one end of the switch S22. The

other of the two electrodes of the capacitor C12 is connected to one end of the switch S31 and one end of the switch S32.

[0087] The capacitor C13 has two electrodes. One of the two electrodes of the capacitor C13 is connected to the one end of the switch S31 and one end of the switch S32. The other of the two electrodes of the capacitor C13 is connected to one end of the switch S41 and one end of the switch S42.

[0088] The capacitor C14 has two electrodes. One of the two electrodes of the capacitor C14 is connected to one end of the switch S13 and one end of the switch S14. The other of the two electrodes of the capacitor C14 is connected to one end of the switch S23 and one end of the switch S24.

[0089] The capacitor C15 has two electrodes. One of the two electrodes of the capacitor C15 is connected to the one end of the switch S23 and the one end of the switch S24. The other of the two electrodes of the capacitor C15 is connected to one end of the switch S33 and one end of the switch S34.

[0090] The capacitor C16 has two electrodes. One of the two electrodes of the capacitor C16 is connected to the one end of the switch S33 and the one end of the switch S34. The other of the two electrodes of the capacitor C16 is connected to one end of the switch S43 and one end of the switch S44.

[0091] Each of a set of the capacitors C11 and C14, a set of the capacitors C12 and C15, and a set of the capacitors C13 and C16 can perform charging and discharging in a complementary manner, with a first phase and a second phase being repeated.

[0092] Specifically, in the first phase, the switches S12, S13, S22, S23, S32, S33, S42, and S43 are turned ON. As a result, for example, the one of the two electrodes of the capacitor C12 is connected to the node N3, the other of the two electrodes of the capacitor C12 and the one of the two electrodes of the capacitor C15 are connected to the node N2, and the other of the two electrodes of the capacitor C15 is connected to the node N1.

[0093] In contrast, in the second phase, the switches S11, S14, S21, S24, S31, S34, S41, and S44 are turned ON. As a result, for example, the one of the two electrodes of the capacitor C15 is connected to the node N3, the other of the two electrodes of the capacitor C15 and the one of the two electrodes of the capacitor C12 are connected to the node N2, and the other of the two electrodes of the capacitor C12 is connected to the node N1.

[0094] Repetition of such first and second phases enables the other of the capacitor C12 and the other of the capacitor C15 to discharge to the capacitor C30, for example, while the one of the capacitor C12 and one of the capacitor C15 are being charged from the node N2. That is, the capacitors C12 and C15 can perform charging and discharging in a complementary manner.

[0095] Similarly to the set of the capacitors C12 and C15, the repetition of the first and second phases enables each of the set of the capacitors C11 and C14 and the set of the capacitors C13 and C16 to perform charging and discharging in a complementary manner.

[0096] Each of the capacitors C10, C20, C30, and C40 is configured to function as a smoothing capacitor in an exemplary aspect. That is, each of the capacitors C10, C20, C30, and C40 is used to hold and smoothen the voltages V1 to V4 at the nodes N1 to N4.

[0097] The capacitor C10 is connected between the node N1 and the ground. Specifically, one of two electrodes of the

Specifically, the one end of the switch S44 is connected to the other of the two electrodes of the capacitor C16. Meanwhile, another end of the switch S44 is connected to the node N1. That is, the other end of the switch S44 is connected to the other end of the switch S31, the other end of the switch S42, and the other end of the switch S33.

[0117] The switches S11, S12, S13, S14, S21, S22, S23, and S24 are each at least one switch included in the switched-capacitor circuit 12.

[0118] A first set of switches including the switches S12, S13, S22, S23, S32, S33, S42, and S43 and a second set of switches including the switches S11, S14, S21, S24, S31, S34, S41, and S44 are switched ON and OFF based on a control signal S2 in a complementary manner. Specifically, in the first phase, the first set of switches is turned ON and the second set of switches is turned OFF. Conversely, in the second phase, the first set of switches is turned OFF and the second set of switches is turned ON.

[0119] For example, in one of the first phase and the second phase, charging of the capacitors C10 to C40 is performed from the capacitors C11 to C13, and in the other of the first phase and the second phase, charging of the capacitors C10 to C40 is performed from the capacitors C14 to C16. That is, the capacitors C10 to C40 are constantly charged from the capacitors C11 to C13 or the capacitors C14 to C16. Therefore, even when electric currents flow from the nodes N1 to N4 to the output switching circuits 13A and 13B at a high speed, it is possible to suppress fluctuations in potentials of the nodes N1 to N4 because the nodes N1 to N4 are replenished with electric charges at a high speed.

[0120] By operating in this manner, the switched-capacitor circuit 12 can maintain an approximately equal voltage at both ends of each of the capacitors C10, C20, C30, and C40. Specifically, the voltages V1 to V4 (voltages relative to the ground potential) that satisfy $V1:V2:V3:V4=1:2:3:4$ are maintained at four nodes labelled V1 to V4. Voltage levels of the voltages V1 to V4 correspond to a plurality of discrete voltage levels that can be supplied by the switched-capacitor circuit 12 to the output switching circuits 13A and 13B.

[0121] It is noted that a voltage ratio (V1:V2:V3:V4) is not limited to (1:2:3:4). For example, the voltage ratio (V1:V2:V3:V4) may be (1:2:4:8) in an exemplary aspect.

[0122] In addition, a configuration of the switched-capacitor circuit 12 illustrated in FIG. 4 is merely an example, and is not limited thereto. In FIG. 4, the switched-capacitor circuit 12 is configured to be able to supply four discrete voltages, but the configuration is not limited thereto. The switched-capacitor circuit 12 may be configured to be able to supply voltages of any number of discrete voltage levels, which is 2 or more. For example, in the case of supplying two discrete voltages, the switched-capacitor circuit 12 should include at least the capacitors C12 and C15 as well as the switches S21 to S24 and S31 to S34.

[2.2 Circuit Configuration of Output Switching Circuits 13A and 13B]

[0123] As illustrated in FIG. 4, the output switching circuit 13A includes input terminals 131A to 134A, switches S51A to S54A, and an output terminal 130A.

[0124] The output terminal 130A is connected to an input terminal 140A of the filter circuit 14A. The output terminal

130A is a terminal for supplying a power supply voltage selected from the voltages V1 to V4 to the PA module 2A via the filter circuit 14A.

[0125] The input terminals 131A to 134A are connected to the nodes N4 to N1 of the switched-capacitor circuit 12, respectively. The input terminals 131A to 134A are terminals for receiving the voltages V4 to V1 from the switched-capacitor circuit 12.

[0126] The switch S51A is connected between the input terminal 131A and the output terminal 130A. Specifically, the switch S51A has a terminal connected to the input terminal 131A and a terminal connected to the output terminal 130A. In this connection configuration, by being switched ON and OFF by a control signal S3A, the switch S51A can switch between connection and disconnection between the input terminal 131A and the output terminal 130A.

[0127] The switch S52A is connected between an input terminal 132A and the output terminal 130A. Specifically, the switch S52A has a terminal connected to the input terminal 132A and a terminal connected to the output terminal 130A. In this connection configuration, by being switched ON and OFF by a control signal S3A, the switch S52A can switch between connection and disconnection between the input terminal 132A and the output terminal 130A.

[0128] The switch S53A is connected between an input terminal 133A and the output terminal 130A. Specifically, the switch S53A has a terminal connected to the input terminal 133A and a terminal connected to the output terminal 130A. In this connection configuration, by being switched ON and OFF by a control signal S3A, the switch S53A can switch between connection and disconnection between the input terminal 133A and the output terminal 130A.

[0129] The switch S54A is connected between the input terminal 134A and the output terminal 130A. Specifically, the switch S54A has a terminal connected to the input terminal 134A and a terminal connected to the output terminal 130A. In this connection configuration, by being switched ON and OFF by a control signal S3A, the switch S54A can switch between connection and disconnection between the input terminal 134A and the output terminal 130A.

[0130] The switches S51A and S52A are each at least one switch included in the output switching circuit 13A.

[0131] These switches S51A to S54A are controlled to be exclusively ON. That is, only one of the switches S51A to S54A is turned ON, and remaining switches of the switches S51A to S54A are turned OFF. This allows the output switching circuit 13A to output one voltage selected from the voltages V1 to V4.

[0132] As illustrated in FIG. 4, the output switching circuit 13B includes input terminals 131B to 134B, switches S51B to S54B, and an output terminal 130B.

[0133] The output terminal 130B is connected to an input terminal 140B of the filter circuit 14B. The output terminal 130B is a terminal for supplying a power supply voltage selected from the voltages V1 to V4 to the PA module 2B via the filter circuit 14B.

[0134] The input terminals 131B to 134B are connected to the nodes N4 to N1 of the switched-capacitor circuit 12,

respectively. The input terminals 131B to 134B are terminals for receiving the voltages V4 to V1 from the switched-capacitor circuit 12.

[0135] Since configurations of the switches S51B and S54B are similar to those of the switches S51A to S54A, a description thereof is omitted here.

[0136] The switches S51B and S52B are each at least one switch included in the output switching circuit 13B.

[0137] These switches S51B to S54B are controlled to be exclusively ON. That is, only one of the switches S51B to S54B is turned ON, and remaining switches of the switches S51B to S54B are turned OFF. This allows the output switching circuit 13B to output one voltage selected from the voltages V1 to V4.

[0138] It is noted that configurations of the output switching circuits 13A and 13B illustrated in FIG. 4 are merely examples, and are not limited thereto. In particular, the switches S51A to S54A may have any configuration as long as they can selectively connect at least one of the four input terminals 131A to 134A to the output terminal 130A. In addition, the switches S51B to S54B may have any configuration as long as they can selectively connect at least one of the four input terminals 131B to 134B to the output terminal 130B. For example, the output switching circuit 13A may further include a switch that is connected between the switches S51A to S53A and the switches S54A and the output terminal 130A. In addition, for example, the output switching circuit 13B may further include a switch connected between the switches S51B to S53B and the switch S54B and the output terminal 130B. In addition, for example, the output switching circuit 13A may further include a switch between the switches S51A and S52A and the switches S53A and S54 and the output terminal 130A. In addition, for example, the output switching circuit 13B may further include a switch between the switches S51B and S52B and the switches S53B and S54B and the output terminal 130B.

[0139] It is noted that when two discrete voltages are supplied from the switched-capacitor circuit 12 according to an exemplary aspect, the output switching circuit 13A may include at least two of the switches S51A to S54A, and the output switching circuit 13B should include at least two of the switches S51B to S54B.

[2.3 Circuit Configuration of Pre-Regulator Circuit 11]

[0140] As illustrated in FIG. 4, the pre-regulator circuit 11 includes an input terminal 110, output terminals 111 to 114, inductor connection terminals 115 and 116, switches S61 to S63, as well as S71 and S72, a power inductor L71, and capacitors C61 to C64.

[0141] The input terminal 110 is a direct current voltage input terminal for receiving an input voltage from a direct-current power supply.

[0142] The output terminal 111 is an output terminal for the voltage V4. That is, the output terminal 111 is a terminal for supplying the voltage V4 to the switched-capacitor circuit 12. The output terminal 111 is connected to the node N4 of the switched-capacitor circuit 12.

[0143] The output terminal 112 is an output terminal for the voltage V3. That is, the output terminal 112 is a terminal for supplying the voltage V3 to the switched-capacitor circuit 12. The output terminal 112 is connected to the node N3 of the switched-capacitor circuit 12.

[0144] The output terminal 113 is an output terminal for the voltage V2. That is, the output terminal 113 is a terminal for supplying the voltage V2 to the switched-capacitor circuit 12. The output terminal 113 is connected to the node N2 of the switched-capacitor circuit 12.

[0145] The output terminal 114 is an output terminal for the voltage V1. That is, the output terminal 114 is a terminal for supplying the voltage V1 to the switched-capacitor circuit 12. The output terminal 114 is connected to the node N1 of the switched-capacitor circuit 12.

[0146] The inductor connection terminal 115 is connected to one end of the power inductor L71. The inductor connection terminal 116 is connected to another end of the power inductor L71.

[0147] The switch S71 is connected between the input terminal 110 and the one end of the power inductor L71. Specifically, the switch S71 has a terminal connected to the input terminal 110 and a terminal connected to the one end of the power inductor L71 via the inductor connection terminal 115. In this connection configuration, by switching ON/OFF based on a control signal S1, the switch S71 can switch between connection and disconnection between the input terminal 110 and the one end of the power inductor L71.

[0148] The switch S72 is connected between the one end of the power inductor L71 and the ground. Specifically, the switch S72 has a terminal connected to the one end of the power inductor L71 via the inductor connection terminal 115, and a terminal connected to the ground. In this connection configuration, by switching ON/OFF based on the control signal S1, the switch S72 can switch between connection and disconnection between the one end of the power inductor L71 and the ground.

[0149] The switch S61 is connected between the other end of the power inductor L71 and the output terminal 111. Specifically, the switch S61 has a terminal connected to the other end of the power inductor L71 via the inductor connection terminal 116, and a terminal connected to the output terminal 111. In this connection configuration, by switching ON/OFF based on the control signal S1, the switch S61 can switch between connection and disconnection between the other end of the power inductor L71 and the output terminal 111.

[0150] A switch S62 is connected between the other end of the power inductor L71 and the output terminal 112. Specifically, the switch S62 has a terminal connected to the other end of the power inductor L71 via the inductor connection terminal 116, and a terminal connected to the output terminal 112. In this connection configuration, by switching ON/OFF based on the control signal S1, the switch S62 can switch between connection and disconnection between the other end of the power inductor L71 and the output terminal 112.

[0151] The switch S63 is connected between the other end of the power inductor L71 and the output terminal 113. Specifically, the switch S63 has a terminal connected to the other end of the power inductor L71 via the inductor connection terminal 116, and a terminal connected to the output terminal 113. In this connection configuration, by switching ON/OFF based on the control signal S1, the switch S63 can switch between connection and disconnection between the other end of the power inductor L71 and the output terminal 113.

[0152] The switches S71 and S72 are each at least one switch included in the pre-regulator circuit 11.

[0153] One of two electrodes of the capacitor C61 is connected to the switch S61 and the output terminal 111. The other of the two electrodes of the capacitor C61 is connected to the switch S62, the output terminal 112, and one of two electrodes of the capacitor C62. The capacitor C61 is configured to function as a smoothing capacitor in an exemplary aspect.

[0154] The one of the two electrodes of the capacitor C62 is connected to the switch S62, the output terminal 112, and the other of the two electrodes of the capacitor C61. The other of the two electrodes of the capacitor C62 is connected to the switch S63, the output terminal 113, and one of the two electrodes of the capacitor C63.

[0155] The one of the two electrodes of the capacitor C63 is connected to the switch S63, the output terminal 113, and the other of the two electrodes of the capacitor C62. The other of the two electrodes of the capacitor C63 is connected to the output terminal 114 and one of the two electrodes of the capacitor C64.

[0156] The one of the two electrodes of the capacitor C64 is connected to the output terminal 114 and the other of the two electrodes of the capacitor C63. The other of the two electrodes of the capacitor C64 is connected to the ground.

[0157] The switches S61 to S63 are controlled to be exclusively ON. That is, only one of the switches S61 to S63 is turned ON, and remaining switches of the switches S61 to S63 are turned OFF. By turning ON only one of the switches S61 to S63, the pre-regulator circuit 11 can change a voltage to be supplied to the switched-capacitor circuit 12 at the voltage levels of the voltages V2 to V4.

[0158] The pre-regulator circuit 11 configured in this manner can supply electric charges to the switched-capacitor circuit 12 via at least one of the output terminals 111 to 113.

[0159] It is noted that when an input voltage to the pre-regulator circuit 11 is converted to one voltage according to an exemplary aspect, the pre-regulator circuit 11 should include at least the switches S71 and S72, and the power inductor L71.

[2.4 Circuit Configuration of Filter Circuits 14A and 14B]

[0160] As illustrated in FIG. 4, the filter circuit 14A includes inductors L1 and L2, a capacitor C1, a switch SW1, the input terminal 140A, and the external output terminal 101.

[0161] The input terminal 140A is connected to the output terminal 130A of the output switching circuit 13A. The input terminal 140A is a terminal for receiving a voltage selected from a plurality of discrete voltages by the output switching circuit 13A.

[0162] The external output terminal 101 is an external connection terminal of the ET module 1A and is connected to the PA module 2A. The external output terminal 101 is a terminal for supplying the PA module 2A with the power supply voltage V_{ETA} which is a plurality of discrete voltages that have passed through the filter circuit 14A.

[0163] The inductor L1 is connected between the input terminal 140A and the external output terminal 101. That is, the inductor L1 is disposed in series in a path connecting the input terminal 140A and the external output terminal 101. Specifically, one end of the inductor L1 is connected to the input terminal 140A, and another end of the inductor L1 is connected to the external output terminal 101.

[0164] The inductor L2 is connected between a path connecting between the inductor L1 and the external output terminal 101, and the ground. That is, the inductor L2 is shunt-connected to the path connecting the input terminal 140A and the external output terminal 101. Specifically, one end of the inductor L2 is connected to a node N42 on the path connecting the inductor L1 and the external output terminal 101, and another end of the inductor L2 is connected to the ground via the capacitor C1. It is noted that the inductor L2 may be connected between the capacitor C1 and the ground, and may be omitted from the filter circuit 14A in an exemplary aspect.

[0165] The capacitor C1 is connected between the inductor L2 and the ground. That is, the capacitor C1 is shunt-connected to the path connecting the input terminal 140A and the external output terminal 101. Specifically, one end of the capacitor C1 is connected to the inductor L2 and another end of the capacitor C1 is connected to the ground.

[0166] The switch SW1 is connected between the input terminal 140A and the external output terminal 101 not via the inductor L1. That is, the switch SW1 is disposed in series in a path that bypasses the inductor L1 between the input terminal 140A and the external output terminal 101. Specifically, one end of the switch SW1 is connected to a node N41 on a path connecting the input terminal 140A and the inductor L1, and another end of the switch SW1 is connected to a node N43 on the path connecting the inductor L1 and the external output terminal 101.

[0167] It is noted that in FIG. 4, the switch SW1 and the inductor L1 are connected in parallel, but another circuit element may be inserted in a path of the switch SW1 and/or a path of the inductor L1. For example, an inductor may be connected between the switch SW1 and the node N43 and/or between the switch SW1 and the node N41.

[0168] In addition, in FIG. 4, although the node N43 to which the other end of the switch SW1 is connected is located between the node N42 to which the inductor L2 is connected and the external output terminal 101, a positional relationship between the node N42 and the node N43 is not limited thereto. For example, the node N42 may be located between the node N43 and the external output terminal 101. In addition, for example, a position of the node N42 may be the same as that of the node N43.

[0169] The switch SW1 connected in this manner is switched ON/OFF based on a control signal S4A. This allows the filter circuit 14A to switch ON/OFF a band-stop filter for removing noise from a plurality of discrete voltages.

[0170] ON/OFF of such a band-stop filter can be controlled based on, for example, a channel band width (i.e., modulation band width) of a radio frequency signal. In addition, when the PA module 2A can amplify transmission signals of a plurality of frequency bands, the ON/OFF of the switch SW1 may be controlled based on the frequency bands of the transmission signals amplified by the PA module 2A. It is noted that control of the band-stop filter is not limited to the above as would be appreciated to one skilled in the art.

[0171] As illustrated in FIG. 4, the filter circuit 14B includes inductors L3 and L4, a capacitor C2, a switch SW2, the input terminal 140B, and the external output terminal 104. A configuration of the filter circuit 14B is described by replacing the inductors L1 and L2 in the filter circuit 14A with the inductors L3 and L4, replacing the capacitor C1 in the filter circuit 14A with the capacitor C2, replacing the

switch SW1 in the filter circuit 14A with the switch SW2, replacing the input terminal 140A in the filter circuit 14A with the input terminal 140B, replacing the external output terminal 101 in the filter circuit 14A with the external output terminal 104, and replacing the control signal S4A with a control signal S4B.

[0172] It is noted that the filter circuits 14A and 14B may be low pass filters for removing noise from the plurality of discrete voltages according to an exemplary aspect.

[3. Circuit Configuration of ET Module 1B and Communication Device 5B in Example 2]

[0173] FIG. 5 is a circuit configuration diagram of an ET module 1B and a communication device 5B according to Example 2. The ET module 1B and the communication device 5B according to this Example are examples of the ET module 1 and the communication device 5 according to the present embodiment, and illustrate a specific example of the circuit configuration of the ET module 1.

[0174] The communication device 5B according to this Example includes the ET module 1B, the PA module 2B, the antennae 3A and 3B, and the RFIC 4. The communication device 5B according to this Example differs from the communication device 5 according to the embodiment in the configuration of the ET module 1B. Therefore, in the following, the communication device 5B according to this Example is mainly described with a focus on the configuration of the ET module 1B.

[0175] The ET module 1B is an example of a tracker module and can supply the power supply voltages V_{ETA} and V_{ETB} , the voltage levels of which vary, to the PA modules 2A and 2B, respectively, based on a tracking mode. The analog ET mode and the APT mode can be used as a tracking mode, but the tracking mode is not limited thereto. As illustrated in FIG. 5, the ET module 1B includes a voltage supply circuit 10B, the control circuit 70, the external output terminals 101 and 104, the external input terminals 102 and 105, and the control signal terminal 103. The ET module 1B according to this Example differs from the ET module 1 according to the embodiment in a configuration of the voltage supply circuit 10B. In the following, the ET module 1B according to this Example is described, omitting the same configurations as those of the ET module 1 according to the embodiment and focusing on different configurations.

[0176] The voltage supply circuit 10B includes DC-DC converters 15A and 15B and linear amplifier circuits 16A and 16B.

[0177] Each of the DC-DC converters 15A and 15B has, for example, one power inductor and outputs a direct-current voltage. It is noted that based on a control signal received from the control circuit 70, the DC-DC converters 15A and 15B can change a level of the direct current voltage for each symbol or for average electric power corresponding to the DC-DC converters 15A and 15B.

[0178] Each of the linear amplifier circuits 16A and 16B receives, from the RFIC 4, an envelope signal representative of a power supply voltage waveform based on an envelope value, and linearly amplifies the envelope signal.

[0179] In operation, when the analog ET mode is applied to the ET module 1i, for example, the linear amplifier circuits 16A and 16B are operated, and the power supply voltages V_{ETA} and V_{ETB} , which correspond to an envelope value and the voltage levels of which vary, are generated at the voltage supply circuit 10B. The voltage supply voltages

V_{ETA} is supplied to the PA module 2A via the external output terminal 101, and the voltage supply voltages V_{ETB} is supplied to the PA module 2B via the external output terminal 104. In addition, when the APT mode is applied to the ET module 1B, for example, the linear amplifier circuits 16A and 16B are not operated, and the power supply voltages V_{ETA} and V_{ETB} , which are stabilized, are generated at the voltage supply circuit 10B. The voltage supply voltages V_{ETA} is supplied to the PA module 2A via the external output terminal 101, and the voltage supply voltages V_{ETB} is supplied to the PA module 2B via the external output terminal 104.

[4. Implementation Examples of ET Module 1A According to Example 1]

[0180] Next, as an implementation example of the ET module 1 configured as described above, an implementation configuration of the ET module 1A according to Example 1 is described with reference to FIGS. 6A to 6C.

[0181] It is noted that in this Example, the power inductor L71 included in the pre-regulator circuit 11 is not disposed on a module laminate 90, but is not limited thereto. That is, the power inductor L71 may be disposed on the module laminate 90.

[0182] FIG. 6A is a plan view of the ET module 1A according to Example 1. FIG. 6B is a plan view of the ET module 1A according to Example 1. FIG. 6A is a view of a main surface 90a side of the module laminate 90 viewed from the z-axis positive side. FIG. 6B is a perspective view of a main surface 90b side of the module laminate 90 from the z-axis positive side. FIG. 6C is a cross-sectional view of the ET module 1A according to Example 1. A cross section of the ET module 1A in FIG. 6C is a cross section taken along VIC-VIC line in FIGS. 6A and 6B.

[0183] It is noted that in FIGS. 6A to 6C, illustration of some of wiring lines connecting a plurality of circuit components disposed on the module laminate 90 is omitted. In FIGS. 6A and 6B, illustration of a resin member 91 covering the plurality of circuit components is omitted. Furthermore, in FIG. 6C, illustration of a shield electrode layer covering the resin member 91 is omitted. It is also noted that the resin member 91 and the shield electrode layer may be omitted in an exemplary aspect. In addition, it is noted that hatched blocks in FIG. 6A represent optional circuit components that may be omitted in exemplary aspects.

[0184] As illustrated in FIGS. 6A and 6B, the ET module 1A includes the module laminate 90, an integrated circuit 80, the external output terminals 101 and 104, and the external input terminals 102 and 105.

[0185] The module laminate 90 has the main surfaces 90a and 90b that face each other. A ground plane or the like is formed in the module laminate 90 and on the main surface 90a. It is noted that in FIGS. 6A and 6B, the module laminate 90 has a rectangular shape in plan view, but the shape of the module laminate 90 is not limited thereto as would be appreciated to one skilled in the art.

[0186] The module laminate 90 may be, but is not limited to, for example, a low temperature co-fired ceramics (LTCC: Low Temperature Co-fired Ceramics) board or a high temperature co-fired ceramics (HTCC: High Temperature Co-fired Ceramics) board, which have a laminated structure of a plurality of dielectric layers, a component-embedded board, a board having a redistribution layer (RDL: Redistribution Layer), or a printed circuit board, or the like.

[0187] The module laminate 90 has sides P1 (a first side), P2 (a second side), P3, and P4 that form at least a part of an outer shape of the module laminate 90 in plan view of the module laminate 90. The side P1 is adjacent to the sides P2 and P4. The side P2 is adjacent to the sides P1 and P3. The side P3 is adjacent to the sides P2 and P4. The side P4 is adjacent to the sides P1 and P3.

[0188] The integrated circuit 80 is disposed on the main surface 90a of the module laminate 90, and has a PR switch portion 11S, an SC switch portion 12S, an OS switch portion 13S, an FL switch portion 14S, and a digital control section 70A. The PR switch portion 11S includes the switches S61 to S63, S71, and S72 of the pre-regulator circuit 11. The SC switch portion 12S includes the switches S11 to S14, S21 to S24, S31 to S34, and S41 to S44 of the switched-capacitor circuit 12. The OS switch portion 13S includes the switches S51A to S54A of the output switching circuit 13A and the switches S51B to S54B of the output switching circuit 13B. The FL switch portion 14S includes the switch SW1 of the filter circuit 14A and the switch SW2 of the filter circuit 14B. The digital control section 70A includes the control circuit 70.

[0189] In addition, in FIG. 6A, the integrated circuit 80 has a rectangular shape in plan view of the module laminate 90, but the shape of the integrated circuit 80 is not limited thereto.

[0190] The integrated circuit 80 includes, for example, a CMOS (Complementary Metal Oxide Semiconductor), and, specifically, may be manufactured by an SOI (Silicon on Insulator) process. It is noted that the integrated circuit 80 is not limited to a CMOS.

[0191] It is noted that the integrated circuit 80 is not necessarily configured by, for example, a single chip, and may be configured by a plurality of chips. In addition, the integrated circuit 80 only should include at least the SC switch portion 12S and the OS switch portion 13S, and the PR switch portion 11S, the FL switch portion 14S, and the digital control section 70A may be disposed outside of the integrated circuit 80.

[0192] Furthermore, as illustrated in FIG. 6B, a plurality of external connection terminals 150 is disposed on the main surface 90B. The plurality of external connection terminals 150 is electrically connected to a plurality of electronic components disposed on the main surface 90a via a via conductor or the like formed in the module laminate 90. The plurality of external connection terminals 150 may be a copper electrode but is not limited thereto. For example, a solder electrode may be used as the plurality of external connection terminals 150.

[0193] Each of the external output terminals 101 and 104, and the external input terminals 102 and 105 is one of the plurality of external connection terminals 150. Some of the plurality of external connection terminals 150 other than the external output terminals 101 and 104, and the external input terminals 102 and 105 are set to the ground.

[0194] Here, as illustrated in FIG. 6B, the external output terminal 101 is disposed adjacent to the side P1, and the external output terminal 104 is not disposed adjacent to the side P1, but is disposed adjacent to the side P2.

[0195] As a result of this, as compared to the configuration in which the external output terminals 101 and 104 are adjacently disposed on the same side of the module laminate 90, the wiring lines supplying the power supply voltages V_{ETA} and V_{ETB} can be disposed at a distance from each other.

Therefore, the heat dissipation of the ET module 1A can be improved, and the waveform distortion of the power supply voltages V_{ETA} and V_{ETB} caused by the heat generation can be suppressed, thereby suppressing the efficiency deterioration.

[0196] In addition, the external input terminal 102 is disposed adjacent to the side P1. As a result of this, the external input terminal 102 and the external output terminal 101 are disposed in proximity, so that the total wiring length of the wiring line connecting the external output terminal 101 and the PA module 2A and the wiring line connecting the external input terminal 102 and the PA module 2A can be reduced. Therefore, the waveform distortion of the power supply voltage V_{ETA} output from the ET module 1A can be suppressed, thereby suppressing the efficiency deterioration.

[0197] Furthermore, in the plan view of the module laminate 90, the external output terminal 101 and the external input terminal 102 are adjacently disposed. This configuration further reduces the above total wiring length.

[0198] In addition, the external input terminal 105 is disposed adjacent to the side P2. As a result of this, the external input terminal 105 and the external output terminal 104 are disposed in proximity, so that the total wiring length of the wiring line connecting the external output terminal 104 and the PA module 2B and the wiring line connecting the external input terminal 105 and the PA module 2B can be reduced. Therefore, the waveform distortion of the power supply voltage V_{ETB} output from the ET module 1A can be suppressed, thereby suppressing the efficiency deterioration.

[0199] Furthermore, in the plan view of the module laminate 90, the external output terminal 104 and the external input terminal 105 are adjacently disposed. This configuration further reduces the above total wiring length.

[0200] In addition, as illustrated in FIG. 6B, in the plan view of the module laminate 90, the external connection terminal 150 (GND) set to the ground potential is disposed adjacent to the external output terminal 101, and the external connection terminal 150 (GND) set to the ground potential is disposed adjacent to the external output terminal 104.

[0201] As a result of this, interference between the power supply voltage V_{ETA} supplied via the external output terminal 101 and the power supply voltage V_{ETB} supplied via the external output terminal 104 can be suppressed, thereby suppressing the waveform distortion of the power supply voltage V_{ETA} and the power supply voltage V_{ETB} .

[0202] It is noted that in the plan view of the module laminate 90, the external connection terminal 150 (GND) set to the ground potential may be disposed between the external output terminal 101 and the external output terminal 104.

[0203] As a result of this, the interference between the power supply voltage V_{ETA} supplied via the external output terminal 101 and the power supply voltage V_{ETB} supplied via the external output terminal 104 can be suppressed, thereby suppressing the waveform distortion of the power supply voltage V_{ETA} and the power supply voltage V_{ETB} .

[0204] Furthermore, in addition to the ET module 1A includes the module laminate 90, the integrated circuit 80, and a plurality of the external connection terminals 150, the ET module 1A includes the capacitors C61 to C64 included in the pre-regulator circuit 11, the capacitors C11 to C16 and the capacitors C10 to C40 included in the switched-capacitor circuit 12, the inductors L1 and L2 as well as the capacitor C1 included in the filter circuit 14A, the inductors L3 and L4 as well as the capacitor C2 included in the filter circuit 14B, and the resin member 91.

[0205] The integrated circuit 80, the capacitors C61 to C64, the capacitors C11 to C16, the capacitors C10 to C40, the inductors L1 to L4 as well as the capacitors C1 and C2, and the resin member 91 are disposed on the main surface 90a.

[0206] The resin member 91 covers at least some of the plurality of electronic components on the main surface 90a. The resin member 91 can be configured to ensure reliability such as mechanical strength and moisture resistance or the like of the plurality of electronic components on the main surface 90a. It is noted that the resin member 91 may be omitted from the ET module 1A in an exemplary aspect.

[0207] Each of the capacitors C61 to C64, the capacitors C11 to C16, and the capacitors C10 to C40 is implemented as a chip capacitor. In this aspect, a chip capacitor can refer to a surface mount device (SMD: Surface Mount Device) that forms a capacitor. It is noted that the implementation of a plurality of the capacitors is not limited to a chip capacitor. For example, some or all of the plurality of capacitors may be included in an integrated passive device (IPD: Integrated Passive Device) or may be included in the integrated circuit 80 in various exemplary aspects.

[0208] The inductors L1 to L4 are implemented as chip inductors. A chip inductor can refer to an SMD that forms an inductor. It is noted that the implementation of the inductors L1 to L4 is not limited to a chip inductor. For example, some or all of the inductors L1 to L4 may be included in an IPD in various exemplary aspects.

[0209] The plurality of capacitors and inductors disposed in this manner on the main surface 90a is grouped for each circuit and is disposed around the integrated circuit 80.

[0210] Specifically, a group of the capacitors C61 to C64 included in the pre-regulator circuit 11 is disposed in a region on the main surface 90a sandwiched between a straight line along a left edge of the integrated circuit 80 and a straight line along a left edge of the module laminate 90 in plan view of the module laminate 90. As a result, a group of circuit components included in the pre-regulator circuit 11 is disposed close to the PR switch portion 11S in the integrated circuit 80.

[0211] In plan view of the module laminate 90, a group of the capacitors C11 to C16 and the capacitors C10 to C40 included in the switched-capacitor circuit 12 is disposed in a region on the main surface 90a sandwiched between a straight line along an upper edge of the integrated circuit 80 and a straight line along an upper edge of the module laminate 90, as well as in a region on the main surface 90a sandwiched between a straight line along a right edge of the integrated circuit 80 and a straight line along a right edge of the module laminate 90. As a result, a group of circuit components included in the switched-capacitor circuit 12 is disposed close to the SC switch portion 12S of the integrated circuit 80. That is, the SC switch portion 12S is disposed closer to the capacitors C11 to C16 and the capacitors C10 to C40 of the switched-capacitor circuit 12 than each of the PR switch portion 11S and the OS switch portion 13S is.

[0212] In plan view of the module laminate 90, a group of the inductors L1 to L4 and the capacitors C1 and C2 included in the filter circuits 14A and 14B is disposed in a region on the main surface 90a sandwiched between a straight line along a lower edge of the integrated circuit 80 and a straight line along a lower edge of the module laminate 90. As a result, a group of circuit components included in the filter circuits 14A and 14B is disposed close to the FL switch

portion 14S in the integrated circuit 80. That is, the FL switch portion 14S is disposed closer to the inductors L1 to L4 and the capacitors C1 and C2 of the filter circuits 14A and 14B than each of the PR switch portion 11S and the SC switch portion 12S is.

[0213] It is noted that configurations of the ET module 1A illustrated in FIGS. 6A to 6C are merely examples, and are not limited thereto. For example, some of the capacitors and the inductors disposed on the main surface 90a may be formed within the module laminate 90. In addition, some of the capacitors and the inductors disposed on the main surface 90a may be omitted from the ET module 1A and are not disposed on the module laminate 90.

[5. Arrangement of ET Module 1A and Communication Device 5A According to Example 1]

[0214] Next, a description is given of the arrangement on a motherboard 92 of the communication device 5A according to Example 1 with reference to FIG. 7. FIG. 7 is a plan view of the communication device 5A according to Example 1. As illustrated in FIG. 7, the communication device 5A includes the motherboard 92, the RFIC 4, the ET module 1A, the PA modules 2A and 2B, and the capacitors 31A and 31B. The PA module 2A includes the power amplifiers 41 to 43, and the PA module 2B includes the power amplifiers 44 to 46.

[0215] The motherboard 92 can be, but is not limited to, for example, a printed circuit board, an LTCC board or an HTCC board, which have a laminated structure of a plurality of dielectric layers, a component-embedded board, a board having an RDL, or the like.

[0216] The RFIC 4, the ET module 1A, the PA modules 2A and 2B, and the capacitors 31A and 31B are disposed on a main surface of the motherboard 92.

[0217] The capacitor 31A is disposed on the motherboard 92 between the ET module 1A and the PA module 2A. The capacitor 31A is connected between a node on the first path connecting the external output terminal 101 and the PA module 2A, and the external input terminal 102. The capacitor 31A is implemented as a chip capacitor. It is noted that the capacitor 31A is not limited to a chip capacitor, and may be included in an IPD, or may be formed of a planar electrode in the motherboard 92 in various exemplary aspects.

[0218] The external output terminal 101 is disposed adjacent to the side P1, and the external output terminal 104 is not disposed adjacent to the side P1, but is disposed adjacent to the side P2.

[0219] As a result of this, as compared to the configuration in which the external output terminals 101 and 104 are adjacently disposed on the same side of the module laminate 90, the wiring lines supplying the power supply voltages V_{ETA} and V_{ETB} can be disposed at a distance from each other. Therefore, the heat dissipation of the ET module 1A can be improved, and the waveform distortion of the power supply voltages V_{ETA} and V_{ETB} caused by the heat generation can be suppressed, thereby suppressing the efficiency deterioration.

[0220] In addition, the external input terminal 102 is disposed adjacent to the side P1. As a result of this, the external input terminal 102 and the external output terminal 101 are disposed in proximity, so that the total wiring length of the wiring line connecting the external output terminal 101 and the PA module 2A and the wiring line connecting the external input terminal 102 and the PA module 2A can

be reduced. Therefore, the waveform distortion of the power supply voltage V_{ETA} output from the ET module 1A can be suppressed, thereby suppressing the efficiency deterioration.

[0221] Furthermore, in the plan view of the module laminate 90, the external output terminal 101 and the external input terminal 102 are adjacently disposed. This configuration further reduces the above total wiring length.

[0222] In addition, the external input terminal 105 is disposed adjacent to the side P2. As a result of this, the external input terminal 105 and the external output terminal 104 are disposed in proximity, so that the total wiring length of the wiring line connecting the external output terminal 104 and the PA module 2B and the wiring line connecting the external input terminal 105 and the PA module 2B can be reduced. Therefore, the waveform distortion of the power supply voltage V_{ETB} output from the ET module 1A can be suppressed, thereby suppressing the efficiency deterioration.

[0223] Furthermore, in the plan view of the module laminate 90, the external output terminal 104 and the external input terminal 105 are adjacently disposed. This configuration further reduces the above total wiring length.

[0224] In addition, the external output terminal 101 and the external input terminal 102 are connected to the capacitor 31A. As a result of this configuration, when the APT mode is applied to the ET module 1A, the capacitor 31A can be configured to function as the bypass condenser, and the voltage level of the power supply voltage V_{ETA} can be stabilized.

[0225] In addition, the external output terminal 104 and the external input terminal 105 are connected to the capacitor 31B. As a result of this configuration, when the APT mode is applied to the ET module 1A, the capacitor 31B can be configured to function as the bypass condenser, and the voltage level of the power supply voltage V_{ETB} can be stabilized.

[6. Arrangement of ET module 1B and Communication Device 5B According to Example 2]

[0226] Next, a description is given of the arrangement on the motherboard 92 of the communication device 5B according to Example 2 with reference to FIG. 8. FIG. 8 is a plan view of the communication device 5B according to Example 2. As illustrated in FIG. 8, the communication device 5B includes the motherboard 92, the RFIC 4, the ET module 1i, the PA modules 2A and 2B, and inductors 33A and 33B. The PA module 2A includes the power amplifiers 41 to 43, and the PA module 2B includes the power amplifiers 44 to 46. It is noted that as compared to the circuit configuration of the communication device 5B illustrated in FIG. 5, the inductors 33A and 33B are added to the communication device 5B illustrated in FIG. 8. An implementation configuration of the communication device 5B illustrated in FIG. 8 mainly differs from that of the communication device 5A illustrated in FIG. 7 in that inductors 33A and 33B are disposed instead of the capacitors 31A and 31B. In the following, the implementation configuration of the communication device 5B according to Example 2 is described with a focus on different points from the implementation configuration of the communication device 5A according to Example 1, and a description of the same points is omitted.

[0227] The RFIC 4, the ET module 1B, the PA modules 2A and 2B, and the inductors 33A and 33B are disposed on the main surface of the motherboard 92.

[0228] The inductor 33A is disposed on the motherboard 92 between the ET module 1B and the PA module 2A. The

inductor 33A is disposed in series on the first path connecting the external output terminal 101 and the PA module 2A.

[0229] As a result of this configuration, when the analog ET mode is applied to the ET module 1B, the inductor 33A can be configured to function as a choke coil that blocks a radio frequency signal leaking from the PA module 2A. Therefore, superimposition of radio frequency noise on the power supply voltage V_{ETA} output from the external output terminal 101 can be suppressed.

[0230] The inductor 33B is disposed on the motherboard 92 between the ET module 1B and the PA module 2B. The inductor 33B is disposed in series on the second path connecting the external output terminal 104 and the PA module 2B.

[0231] As a result of this configuration, when the analog ET mode is applied to the ET module 1B, the inductor 33B can be configured to function as a choke coil that blocks a radio frequency signal leaking from the PA module 2B. Therefore, superimposition of radio frequency noise on the power supply voltage V_{ETB} output from the external output terminal 104 can be suppressed.

[0232] It is noted that each of the inductor 33A and 33B is implemented as a chip inductor. However, the inductors 33A and 33B are not limited to chip inductors and may be included in an IPD or may be formed of a planar coil in the motherboard 92 in various exemplary aspects.

[0233] The external input terminal 102 is connected to the voltage supply circuit 10B. This configuration makes it possible to return a feedback signal of the power supply voltage V_{ETA} on the first path to the voltage supply circuit 10B via the external input terminal 102. In addition, the external input terminal 105 is connected to the voltage supply circuit 10B. This makes it possible to return a feedback signal of the power supply voltage V_{ETB} on the second path to the voltage supply circuit 10B via the external input terminal 105. It is noted that the external input terminals 102 and 105 may be connected to the control circuit 70, rather than being connected to the voltage supply circuit 10B.

[0234] It is noted that the inductor 33A may be connected between the external input terminal 102 and a node on the first path. As a result of this configuration, when the analog ET mode is applied to the ET module 1, the inductor 33A can be configured to function as the choke coil that blocks a radio frequency signal leaking from the PA module 2A. Therefore, it is possible to return the feedback signal of the power supply voltage V_{ETA} on the first path to the ET module 1B with high precision via the external input terminal 102.

[0235] In addition, the inductor 33B may be connected between the external input terminal 105 and a node on the second path. As a result of this configuration, when the analog ET mode is applied to the ET module 1B, the inductor 33B can be configured to function as the choke coil that blocks a radio frequency signal leaking from the PA module 2B. Therefore, it is possible to return the feedback signal of the power supply voltage V_{ETB} on the second path to the ET module 1B with high precision via the external input terminal 105.

[7. Arrangement of ET Module 1C and Communication Device 5C According to Modification Example]

[0236] Next, a description is given of arrangement on a motherboard 92 of a communication device 5C according to a modification example with reference to FIG. 9. FIG. 9 is

a plan view of the communication device 5C according to the modification example. As illustrated in FIG. 9, the communication device 5C includes the motherboard 92, the RFIC 4, an ET module 1C, the PA modules 2A, 2B, and 2C, and the capacitors 31A and 31B.

[0237] The ET module 1C includes the module laminate 90, a voltage supply circuit 10C, the control circuit 70, the external output terminals 101, 104, and 106, and the external input terminals 102 and 105, and an external input terminal 107.

[0238] The PA module 2A includes the power amplifiers 41 to 43 and can amplify radio frequency signals in band A to and C that belong to a low band group (600 MHz to 1 GHz). The PA module 2B includes the power amplifiers 44 to 46 and can amplify radio frequency signals in band D to band F that belong to a middle and high band groups (1.5 to 2.8 GHz). The PA module 2C can amplify radio frequency signals in bands that belong to an ultra-high band group (3.3 to 5.0 GHz). The communication device 5C according to this modification example mainly differs from the communication device 5A according to Example 1 in that the PA module 2C is added. In the following, an implementation configuration of the communication device 5C according to this modification example is described with a focus on different points from the implementation configuration of the communication device 5A according to Example 1, and a description of the same points is omitted.

[0239] The RFIC 4, the ET module 1C, the PA modules 2A, 2B, and 2C, and the capacitors 31A and 31B are disposed on a main surface of the motherboard 92.

[0240] The capacitor 31A is disposed on the motherboard 92 between the ET module 1C and the PA module 2A. The capacitor 31A is connected between a node on the first path connecting the external output terminal 101 and the PA module 2A, and the external input terminal 102.

[0241] The capacitor 31B is disposed on the motherboard 92 between the ET module 1C and the PA module 2B. The capacitor 31B is connected between a node on the second path connecting the external output terminal 104 and the PA module 2B, and the external input terminal 105.

[0242] The external output terminal 101 is disposed adjacent to the side P1. The external output terminal 104 is not disposed adjacent to the side P1 but is disposed adjacent to the side P2. The external output terminal 106 is not disposed adjacent to the sides P1 and P2 but is disposed adjacent to the side P3.

[0243] As result of this, as compared to a configuration in which the external output terminals 101, 104, and 106 are adjacently disposed on a same side of the module laminate 90, wiring lines supplying the power supply voltage V_{ET} can be disposed at a distance from each other. Therefore, heat dissipation of the ET module 1C can be improved, and waveform distortion of the power supply voltages V_{ET} caused by the heat generation can be suppressed, thereby suppressing efficiency deterioration.

[0244] In addition, the external input terminal 102 is disposed adjacent to the side P1. As a result of this, the external input terminal 102 and the external output terminal 101 are adjacently disposed in proximity, so that the total wiring length of the wiring line connecting the external output terminal 101 and the PA module 2A and the wiring line connecting the external input terminal 102 and the PA module 2A can be reduced. Therefore, the waveform dis-

tortion of the power supply voltage V_{ET} output from the ET module 1C can be suppressed, thereby suppressing the efficiency deterioration.

[0245] Furthermore, in the plan view of the module laminate 90, the external output terminal 101 and the external input terminal 102 are adjacently disposed. This configuration further reduces the above total wiring length.

[0246] In addition, the external input terminal 105 is disposed adjacent to the side P2. As a result of this, the external input terminal 105 and the external output terminal 104 are disposed in proximity, so that the total wiring length of the wiring line connecting the external output terminal 104 and the PA module 2B and the wiring line connecting the external input terminal 105 and the PA module 2B can be reduced. Therefore, the waveform distortion of the power supply voltage V_{ET} output from the ET module 1C can be suppressed, thereby suppressing the efficiency deterioration.

[0247] Furthermore, in the plan view of the module laminate 90, the external output terminal 104 and the external input terminal 105 are adjacently disposed. This configuration further reduces the above total wiring length.

[0248] In addition, the external input terminal 107 is disposed adjacent to the side P3. As a result of this, the external input terminal 107 and the external output terminal 106 are disposed in proximity, so that the total wiring length of a wiring line connecting the external output terminal 106 and the PA module 2C and a wiring line connecting the external input terminal 107 and the PA module 2C can be reduced. Therefore, the waveform distortion of the power supply voltage V_{ET} output from the ET module 1C can be suppressed, thereby suppressing the efficiency deterioration.

[0249] Furthermore, in the plan view of the module laminate 90, the external output terminal 106 and the external input terminal 107 are adjacently disposed. This configuration further reduces the above total wiring length.

[0250] In addition, the external output terminal 101 and the external input terminal 102 are connected to the capacitor 31A. As a result of this configuration, when the APT mode is applied to the ET module 1C, the capacitor 31A can be configured to function as a bypass condenser, and the voltage level of the power supply voltage V_{ET} can be stabilized.

[0251] In addition, the external output terminal 104 and the external input terminal 105 are connected to the capacitor 31B. As a result of this configuration, when the APT mode is applied to the ET module 1C, the capacitor 31B can be configured to function as a bypass condenser, and the voltage level of the power supply voltage V_{ET} can be stabilized.

[8. Technical Effects]

[0252] As described above, the ET module 1 according to the embodiment includes the module laminate 90; the external output terminals 101 and 104 disposed on the module laminate 90; and the voltage supply circuit 10 disposed on the module laminate 90 and configured to supply the power supply voltage V_{ETA} based on an envelope signal to the PA module 2A via the external output terminal 101, and to simultaneously supply the power supply voltage V_{ETB} based on an envelope signal to the PA module 2B via the external output terminal 104. The module laminate 90 has the side P1 and the side P2 adjacent to the side P1, the sides P1 and P2 forming at least a part of the outer shape of the module laminate 90 in plan view of the module laminate 90. The

external output terminal **101** is disposed adjacent to the side P1, and the external output terminal **104** is not disposed adjacent to the side P1, but is disposed adjacent to the side P2.

[0253] As a result of this, as compared to the configuration in which the external output terminals **101** and **104** are adjacently disposed on the same side of the module laminate **90**, the wiring lines supplying the power supply voltages V_{ETA} and V_{ETB} can be disposed at a distance from each other. Therefore, the heat dissipation of the ET module **1** can be improved, and the waveform distortion of the power supply voltages V_{ETA} and V_{ETB} caused by the heat generation can be suppressed, thereby suppressing the efficiency deterioration.

[0254] In addition, the ET module **1A** according to Example 1 includes the module laminate **90**; the external output terminals **101** and **104** disposed on the module laminate **90**; and the voltage supply circuit **10A** including at least one integrated circuit **80** disposed on the module laminate **90**. The voltage supply circuit **10A** includes at least one switch included in the switched-capacitor circuit **12**, at least one switch included in the output switching circuit **13A**, and at least one switch included in the output switching circuit **13B**. The switched-capacitor circuit **12** is configured to generate a plurality of discrete voltages based on an input voltage and to output the plurality of discrete voltages generated to the output switching circuits **13A** and **13B**. The output switching circuit **13A** is configured to selectively output at least one of the plurality of discrete voltages generated to the PA module **2A** via the external output terminal **101**, and the output switching circuit **13B** is configured to selectively output at least one of the plurality of discrete voltages generated to the PA module **2B** via the external output terminal **104**. The module laminate **90** has the side P1 and the side P2 adjacent to the side P1, the sides P1 and P2 forming at least a part of the outer shape of the module laminate **90** in plan view of the module laminate **90**. The external output terminal **101** is disposed adjacent to the side P1, and the external output terminal **104** is not disposed adjacent to the side P1, but is disposed adjacent to the side P2.

[0255] As a result of this, as compared to the configuration in which the external output terminals **101** and **104** are adjacently disposed on the same side of the module laminate **90**, the wiring lines supplying the power supply voltages V_{ETA} and V_{ETB} can be disposed at a distance from each other. Therefore, the heat dissipation of the ET module **1A** can be improved, and the waveform distortion of the power supply voltages V_{ETA} and V_{ETB} caused by the heat generation can be suppressed, thereby suppressing the efficiency deterioration.

[0256] In addition, for example, the ET modules **1**, **1A**, **1**, and **1C** further include the external input terminal **102** connected to the first path connecting the external output terminal **101** and the PA module **2A**, and the external input terminal **102** is disposed adjacent to the side P1.

[0257] As a result of this, the external input terminal **102** and the external output terminal **101** are disposed in proximity, so that the total wiring length of the wiring line connecting the external output terminal **101** and the PA module **2A** and the wiring line connecting the external input terminal **102** and the PA module **2A** can be reduced. Therefore, the waveform distortion of the power supply voltage V_{ETA} output from the ET module can be suppressed, thereby suppressing the efficiency deterioration.

[0258] In addition, for example, in the ET modules **1**, **1A**, **1**, and **1C**, the external output terminal **101** and the external input terminal **102** are adjacently disposed in the plan view of the module laminate **90**.

[0259] This configuration further reduces the above total wiring length.

[0260] In addition, for example, the ET modules **1**, **1A**, **1B**, and **1C** further include the external input terminal **105** connected to the second path connecting the external output terminal **104** and the PA module **2B**, and the external input terminal **105** is disposed adjacent to the side P2.

[0261] As a result of this, the external input terminal **105** and the external output terminal **104** are disposed in proximity, so that the total wiring length of the wiring line connecting the external output terminal **104** and the PA module **2B** and the wiring line connecting the external input terminal **105** and the PA module **2B** can be reduced. Therefore, the waveform distortion of the power supply voltage V_{ETB} output from the ET module can be suppressed, thereby suppressing the efficiency deterioration.

[0262] Furthermore, for example, in the plan view of the module laminate **90**, the external output terminal **104** and the external input terminal **105** are adjacently disposed.

[0263] This configuration further reduces the above total wiring length.

[0264] In addition, for example, in the ET modules **1**, **1A**, **1**, and **1C**, in the plan view of the module laminate **90**, the external connection terminal **150** (GND) is disposed adjacent to the external output terminal **101**, and the external connection terminal **150** (GND) is disposed adjacent to the external output terminal **104**.

[0265] As a result of this, interference between the power supply voltage V_{ETA} supplied via the external output terminal **101** and the power supply voltage V_{ETB} supplied via the external output terminal **104** can be suppressed, thereby suppressing the waveform distortion of the power supply voltage V_{ETA} and the power supply voltage V_{ETB} .

[0266] In addition, for example, in the ET modules **1**, **1A**, **1B**, and **1C**, in the plan view of the module laminate **90**, the external connection terminal **150** (GND) may be disposed between the external output terminal **101** and the external output terminal **104**.

[0267] As a result of this, the interference between the power supply voltage V_{ETA} supplied via the external output terminal **101** and the power supply voltage V_{ETB} supplied via the external output terminal **104** can be suppressed, thereby suppressing the waveform distortion of the power supply voltage V_{ETA} and the power supply voltage V_{ETB} .

[0268] In addition, for example, the ET modules **1A** and **1C** may further include a low pass filter connected between the output switching circuit **13A** and the external output terminal **101**.

[0269] This configuration removes noise due to pulse waveforms of the plurality of discrete voltages output from the output switching circuit **13A**.

[0270] In addition, for example, in the ET modules **1A** and **1C**, the external output terminal **101** and the external input terminal **102** are connected to the capacitor **31A**.

[0271] As a result of this configuration, when the APT mode is applied to the ET modules **1A** and **1C**, the capacitor **31A** can be configured to function as the bypass condenser, and the voltage level of the power supply voltage V_{ETA} can be stabilized.

[0272] In addition, for example, the ET modules 1A and 1C further includes the switch 21A that switches between connection and disconnection between the external input terminal 102 and the ground.

[0273] As a result of this configuration, when the digital ET mode is applied to the ET modules 1A and 1C, the switch 21A is in a non-conductive state, so that a radio frequency signal of the wide channel band width (wide modulation band width) can be amplified highly efficiently in the PA module 2A. In addition, when the APT mode is applied to the ET modules 1A and 1C, the switch 21A is in a conductive state, so that the capacitor 31A can be configured to function as the bypass condenser, and the voltage level of the power supply voltage V_{ETA} can be stabilized.

[0274] In addition, for example, in the ET module 1, at least one of the external input terminal 102 and the external output terminal 101 is connected to the inductor 33A.

[0275] As a result of this configuration, when the analog ET mode is applied to the ET module 1i, the inductor 33A can be configured to function as the choke coil that blocks a radio frequency signal leaking from the PA module 2A. Therefore, it is possible to return the feedback signal of the power supply voltage V_{ETA} on the first path to the ET module 1B with high precision via the external input terminal 102. In addition, the superimposition of the radio frequency noise on the power supply voltage V_{ETA} output from the external output terminal 101 can be suppressed.

[0276] In addition, for example, in the ET module 1B, the external input terminal 102 is connected to the voltage supply circuit 10B.

[0277] This makes it possible to return the feedback signal of the power supply voltage V_{ETA} on the first path to the voltage supply circuit 10B via the external input terminal 102.

[0278] In addition, the communication device 5 according to the embodiment includes the RFIC 4 that processes radio frequency signals, the ET module 1 connected to the RFIC 4, the PA modules 2A and 2B, and the motherboard 92 on which the RFIC 4, the ET module 1, as well as the PA modules 2A and 2B are disposed.

[0279] This configuration realizes the effects of the ET module 1 in the communication device 5.

[0280] In addition, for example, the communication device 5A according to Example 1 (the communication device 5C according to a modification example) further includes the capacitor 31A that is disposed on the motherboard 92 and connected to the external output terminal 101.

[0281] As a result of this configuration, when the APT mode is applied to the ET module 1A and 1C, the capacitor 31A can be configured to function as the bypass condenser, and the voltage level of the power supply voltage V_{ETA} can be stabilized.

[0282] In addition, for example, the communication device 5B according to Example 2 further includes the inductor 33A that is disposed on the motherboard 92 and connected to the external output terminal 101.

[0283] As a result of this configuration, when the analog ET mode is applied to the ET module 1i, the inductor 33A can be configured to function as the choke coil that blocks a radio frequency signal leaking from the PA module 2A. Therefore, the superimposition of the radio frequency noise on the power supply voltage V_{ETA} output from the external output terminal 101 can be suppressed.

Additional Exemplary Embodiment

[0284] As described above, the tracker modules and the communication devices according to the present disclosure have been described based on the embodiments, examples, and modification examples. However, the tracker modules and the communication devices according to the exemplary aspects of the present disclosure are not limited to the above-described embodiments, examples, and modification examples. It should be appreciated that the present disclosure also includes other embodiments realized by combining any of the components in the above-described embodiments, examples, and modification examples, modification examples obtained by applying various modifications to the above-described embodiments, examples, and modification examples without departing from the spirit of the exemplary aspects of the present disclosure, or various types of devices that incorporate the tracker modules or the communication devices described above.

[0285] For example, in the circuit configurations of the various circuits according to the above-described embodiment, examples, and modification examples, another circuit element and wiring line, or the like may be inserted between the paths connecting the circuit elements and the signal paths disclosed in the drawings.

[0286] In the following, a description is given of the characteristics of the tracker modules and the communication devices described based on the embodiment described above.

[0287] <1> A tracker module is provided that includes a module laminate; a first external output terminal and a second external output terminal disposed on the module laminate; and a voltage supply circuit disposed on the module laminate and configured to supply a first voltage based on an envelope signal to a first power amplifier via the first external output terminal, and to simultaneously supply a second voltage based on an envelope signal to a second power amplifier via the second external output terminal. The module laminate has a first side and a second side adjacent to the first side. Moreover, the first side and the second side form at least a part of an outer shape of the module laminate in plan view of the module laminate. The first external output terminal is disposed adjacent to the first side, and the second external output terminal is disposed adjacent to the second side but not the first side.

[0288] <2> A tracker module is provided that includes a module laminate; a first external output terminal and a second external output terminal disposed on the module laminate; and a voltage supply circuit including at least one integrated circuit disposed on the module laminate. The at least one integrated circuit includes at least one switch included in a switched-capacitor circuit, at least one switch included in a first output switching circuit, and at least one switch included in a second output switching circuit. The switched-capacitor circuit is configured to generate a plurality of discrete voltages based on an input voltage and to output the generated plurality of discrete voltages to the first output switching circuit and the second output switching circuit. The first output switching circuit is configured to selectively output at least one discrete voltage of the generated plurality of discrete voltages to a first power amplifier via the first external output terminal, and the second output switching circuit is configured to selectively output at least one discrete voltage of the generated plurality of discrete voltages to a second power amplifier via the second external

output terminal. The module laminate has a first side and a second side adjacent to the first side. Moreover, the first side and the second side form at least a part of an outer shape of the module laminate in plan view of the module laminate. The first external output terminal is disposed adjacent to the first side, and the second external output terminal is disposed adjacent to the second side but not the first side.

[0289] <3> The tracker module according to <1> or <2>, further including a first external input terminal connected to a first path connecting the first external output terminal and the first power amplifier, in which the first external input terminal is disposed adjacent to the first side.

[0290] <4> The tracker module according to <3>, wherein, in the plan view of the module laminate, the first external output terminal and the first external input terminal are disposed adjacent to each other.

[0291] <5> The tracker module according to any one of <1> to <3>, further including a second external input terminal connected to a second path connecting the second external output terminal and the second power amplifier, in which the second external input terminal is disposed adjacent to the second side.

[0292] <6> The tracker module according to <5>, wherein, in the plan view of the module laminate, the second external output terminal and the second external input terminal are disposed adjacent to each other.

[0293] <7> The tracker module according to any one of <1> to <6>, wherein, in the plan view of the module laminate, a ground terminal is disposed adjacent to the first external output terminal, and a ground terminal is disposed adjacent to the second external output terminal.

[0294] <8> The tracker module according to any one of <1> to <6>, wherein, in the plan view of the module laminate, a ground terminal is disposed between the first external output terminal and the second external output terminal.

[0295] <9> The tracker module according to <2>, further including a low pass filter connected between the first output switching circuit and the first external output terminal.

[0296] <10> The tracker module according to <3>, in which the first external output terminal and the first external input terminal are connected to a capacitor.

[0297] <11> The tracker module according to <10>, further including a first switch circuit that switches between connection and disconnection between the first external input terminal and a ground.

[0298] <12> The tracker module according to <3>, in which at least one of the first external input terminal and the first external output terminal is connected to an inductor.

[0299] <13> The tracker module according to <3> or <12>, in which the first external input terminal is connected to the voltage supply circuit.

[0300] <14> A communication device is provided that includes a signal processing circuit that processes radio frequency signals; the tracker module according to any one of <1> to <13>, which is connected to the signal processing circuit; the first power amplifier and the second power amplifier; and a motherboard on which the signal processing circuit, the tracker module, and the first power amplifier and the second power amplifier are disposed.

[0301] <15> The communication device according to <14>, further including a capacitor disposed on the motherboard and connected to the first external output terminal.

[0302] <16> The communication device according to <14>, further including an inductor disposed on the motherboard and connected to the first external output terminal. [0303] The exemplary aspects of the present disclosure can be widely used in communication devices, such as mobile phones, as a tracker module that supplies voltages to a power amplifier.

REFERENCE SIGNS LIST

[0304]	1, 1A, 1B, 1C ET MODULE
[0305]	2A, 2B, 2C PA MODULE
[0306]	3A, 3B, 3C ANTENNA
[0307]	4 RFIC
[0308]	5, 5A, 5B, 5C COMMUNICATION DEVICE
[0309]	10, 10A, 10B, 10C VOLTAGE SUPPLY CIRCUIT
[0310]	11 PRE-REGULATOR CIRCUIT
[0311]	11S PR SWITCH PORTION
[0312]	12 SWITCHED-CAPACITOR CIRCUIT
[0313]	12S SC SWITCH PORTION
[0314]	13A, 13B OUTPUT SWITCHING CIRCUIT
[0315]	13S OS SWITCH PORTION
[0316]	14A, 14B FILTER CIRCUIT
[0317]	14S FL SWITCH PORTION
[0318]	15A, 15B DC-DC CONVERTER
[0319]	16A, 16B LINEAR AMPLIFIER CIRCUIT
[0320]	21A, 21B, 61, 62 SWITCH
[0321]	31A, 31B CAPACITOR
[0322]	33A, 33B INDUCTOR
[0323]	41, 42, 43, 44, 45, 46 POWER AMPLIFIER
[0324]	51, 52, 53, 54, 55, 56 FILTER
[0325]	70 CONTROL CIRCUIT
[0326]	70A DIGITAL CONTROL SECTION
[0327]	80 INTEGRATED CIRCUITS
[0328]	90 MODULE LAMINATE
[0329]	90a, 90b MAIN SURFACE
[0330]	91 RESIN MEMBER
[0331]	92 MOTHERBOARD
[0332]	101, 104, 106 EXTERNAL OUTPUT TERMINAL
[0333]	102, 105, 107 EXTERNAL INPUT TERMINALS
[0334]	103 CONTROL SIGNAL TERMINAL
[0335]	110, 131A, 131B, 132A, 132B, 133A, 133B, 134A, 134B, 140A, 140B INPUT TERMINALS
[0336]	111, 112, 113, 114, 130A, 130B OUTPUT TERMINALS
[0337]	150 EXTERNAL CONNECTION TERMINAL
[0338]	P1, P2, P3, P4 SIDE

1. A tracker module comprising:

a module laminate;

a first external output terminal and a second external output terminal on the module laminate; and

a voltage supply circuit on the module laminate and configured to supply a first voltage based on an envelope signal to a first power amplifier via the first external output terminal, and to simultaneously supply a second voltage based on an envelope signal to a second power amplifier via the second external output terminal,

wherein the module laminate has a first side and a second side adjacent to the first side,

wherein the first side and the second side form at least a part of an outer shape of the module laminate in a plan view of the module laminate,

wherein the first external output terminal is disposed adjacent to the first side, and

wherein the second external output terminal is disposed adjacent to the second side but is not disposed adjacent to the first side.

2. The tracker module according to claim 1, further comprising:

a first external input terminal connected to a first path that connects the first external output terminal to the first power amplifier,

wherein the first external input terminal is disposed adjacent to the first side.

3. The tracker module according to claim 2, wherein, in the plan view of the module laminate, the first external output terminal is adjacent to the first external input terminal.

4. The tracker module according to claim 1, further comprising:

a second external input terminal connected to a second path that connects the second external output terminal to the second power amplifier,

wherein the second external input terminal is disposed adjacent to the second side.

5. The tracker module according to claim 4, wherein, in the plan view of the module laminate, the second external output terminal is adjacent to the second external input terminal.

6. The tracker module according to claim 1, wherein, in the plan view of the module laminate, a ground terminal is disposed adjacent to the first external output terminal, and a ground terminal is disposed adjacent to the second external output terminal.

7. The tracker module according to claim 1, wherein in the plan view of the module laminate, a ground terminal is disposed between the first external output terminal and the second external output terminal.

8. The tracker module according to claim 2, wherein the first external output terminal and the first external input terminal are connected to a capacitor.

9. The tracker module according to claim 8, further comprising a first switch circuit configured to switch between a connection and a disconnection between the first external input terminal and a ground.

10. The tracker module according to claim 2, wherein at least one of the first external input terminal and the first external output terminal is connected to an inductor.

11. The tracker module according to claim 2, wherein the first external input terminal is connected to the voltage supply circuit.

12. A tracker module comprising:

a module laminate;

a first external output terminal and a second external output terminal disposed on the module laminate; and

a voltage supply circuit including at least one integrated circuit on the module laminate,

wherein the at least one integrated circuit includes at least one switch included in a switched-capacitor circuit, at least one switch included in a first output switching circuit, and at least one switch included in a second output switching circuit,

wherein the switched-capacitor circuit is configured to generate a plurality of discrete voltages based on an input voltage and to output the generated plurality of discrete voltages to the first output switching circuit and the second output switching circuit,

wherein the first output switching circuit is configured to selectively output at least one discrete voltage of the plurality of discrete voltages to a first power amplifier via the first external output terminal,

wherein the second output switching circuit is configured to selectively output at least one discrete voltage of the plurality of discrete voltages to a second power amplifier via the second external output terminal,

wherein the module laminate has a first side and a second side adjacent to the first side,

wherein the first side and the second side form at least a part of an outer shape of the module laminate in a plan view of the module laminate,

wherein the first external output terminal is disposed adjacent to the first side, and

wherein the second external output terminal is disposed adjacent to the second side but is not disposed adjacent to the first side.

13. The tracker module according to claim 12, further comprising a low pass filter connected between the first output switching circuit and the first external output terminal.

14. The tracker module according to claim 12, further comprising:

a first external input terminal connected to a first path that connects the first external output terminal to the first power amplifier,

wherein the first external input terminal is disposed adjacent to the first side.

15. The tracker module according to claim 14, wherein, in the plan view of the module laminate, the first external output terminal is adjacent to the first external input terminal.

16. The tracker module according to claim 12, further comprising:

a second external input terminal connected to a second path that connects the second external output terminal to the second power amplifier,

wherein the second external input terminal is disposed adjacent to the second side.

17. The tracker module according to claim 16, wherein, in the plan view of the module laminate, the second external output terminal is adjacent to the second external input terminal.

18. A communication device comprising:

a signal processing circuit configured to process radio frequency signals;

the tracker module according to claim 1, the tracker module connected to the signal processing circuit;

the first power amplifier and the second power amplifier; and

a motherboard on which the signal processing circuit, the tracker module, and the first power amplifier and the second power amplifier are disposed.

19. The communication device according to claim 18, further comprising a capacitor disposed on the motherboard and connected to the first external output terminal.

20. The communication device according to claim **18**, further comprising an inductor disposed on the motherboard and connected to the first external output terminal.

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